Lecture Notes on Optimization and Optimal Control

Module 3: Basic Convex Optimization

Quanyan Zhu

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This chapter provides a systematic introduction to convex optimization and convexification techniques, emphasizing the geometric, analytical, and duality structures that make convex problems tractable. We review the foundational concepts of convex sets and convex functions, and explain why convex programs admit strong global optimality guarantees, well-behaved solution structures, and efficient algorithms.

A central theme of the chapter is *convexification*: the transformation of nonconvex optimization problems into convex ones through reformulation, relaxation, or duality. We present exact convex reformulations based on epigraph methods, change of variables, eigenvalue representations, and fractional programming transformations, as well as convex relaxations obtained via lifting, rank relaxation, convex envelopes, and Lagrangian duality. The distinction between exact reformulations and relaxations is emphasized throughout.

The chapter also develops saddle-point characterizations of optimality, Slater's condition, and the Karush–Kuhn–Tucker (KKT) conditions, highlighting their geometric interpretation and their role in duality theory. Classical examples, including water-filling and eigenvalue optimization, illustrate how abstract convex-analytic principles lead to concrete and interpretable solutions in communications, control, and resource allocation.

Together, these topics provide a unified framework for understanding why convex optimization serves as a cornerstone of modern optimization theory and how convexification enables the principled treatment of a broad class of otherwise intractable nonconvex problems.

1 Convex Optimization: Basic Concepts and Geometry

Convex optimization problems constitute a special and highly structured class of optimization problems in which geometry plays a central role. A generic convex optimization problem can be written in the form

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \qquad i \in \mathcal{Q} := \{1, 2, \ldots, q\}, \end{aligned} \tag{1.1}$$

where the objective function $f : \mathbb{R}^n \to \mathbb{R}$ and the constraint functions $g_i : \mathbb{R}^n \to \mathbb{R}$ are assumed to be convex.

The theory of convex optimization is built upon three tightly interconnected concepts: convex sets, which describe the geometry of feasible regions; convex functions, which govern the structure of objective and constraint functions; and convex programs, which combine these elements into optimization problems with remarkable analytical and algorithmic properties. Among these, the geometry of convex sets provides the most immediate intuition for why convex optimization problems admit strong global optimality guarantees and well-behaved solution structures.

In the formulation (1.1), we present the problem using only inequality constraints. Equality constraints can always be represented as two inequality constraints of the form $h(x) \leq 0$ and $-h(x) \leq 0$, although doing so may obscure important geometric properties of the feasible set. In practice, equality constraints are often treated separately, and in some cases they appear implicitly or are disguised as inequality constraints that are always active at feasible points. Understanding this distinction is essential when formulating constraint qualifications and optimality conditions in convex optimization.

1.1 Convex Sets

Convex sets provide the geometric foundation of convex optimization. Their simple defining property leads to strong closure properties and powerful separation results that underpin duality, optimality conditions, and algorithmic design.

Definition 1.1 (Convex Set)**.** A set $C \subseteq \mathbb{R}^n$ is said to be *convex* if for any $x, y \in C$ and any scalar $\lambda \in [0, 1]$, the point

$$z := \lambda x + (1 - \lambda) y$$

also belongs to C.

Geometrically, convexity means that the entire line segment connecting any two points in C lies completely inside C. This seemingly simple property is the source of many of the favorable analytical and computational properties of convex optimization problems.

1.1.1 Examples of Convex Sets

We now present several canonical examples of convex sets that arise frequently in optimization and related fields. Each example illustrates convexity through direct verification from the definition.

Example 1.2 (Affine Equality Constraints)**.** Consider the set

$$F := \{x \in \mathbb{R}^n : Ax = b\},$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

Proposition 1.3. *The set F is convex.*

Proof. Let $x, y \in F$ and $\lambda \in [0, 1]$. Define

$$z := \lambda x + (1 - \lambda) y.$$

By linearity of matrix multiplication,

$$Az = \lambda Ax + (1 - \lambda) Ay = \lambda b + (1 - \lambda) b = b.$$

Hence $z \in F$, which proves that F is convex. □

Example 1.4 (Half-Spaces)**.** A half-space is a set of the form

$$H := \{x \in \mathbb{R}^n : a^\top x \leq b\},$$

where $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$.

Proposition 1.5. *Every half-space is convex.*

Proof. Let $x, y \in H$ and $\lambda \in [0, 1]$. Then

$$a^\top (\lambda x + (1 - \lambda) y) = \lambda a^\top x + (1 - \lambda) a^\top y \leq \lambda b + (1 - \lambda) b = b.$$

Thus $\lambda x + (1-\lambda)y \in H$, proving convexity. $\square$

Half-spaces serve as the fundamental building blocks of polyhedral sets and play a central role in linear and convex programming.

Example 1.6 (Euclidean Balls and Ellipsoids)**.** The Euclidean ball of radius $r > 0$ centered at x_0 is defined by

$$B(x_0, r) := \{x \in \mathbb{R}^n : \|x - x_0\|_2 \leq r\}.$$

Proposition 1.7. *Euclidean balls are convex.*

Proof. Let $x, y \in B(x_0, r)$ and $\lambda \in [0, 1]$. By the triangle inequality,

$$\|\lambda(x - x_0) + (1-\lambda)(y - x_0)\|_2 \leq \lambda\|x - x_0\|_2 + (1-\lambda)\|y - x_0\|_2 \leq r.$$

Therefore, $\lambda x + (1-\lambda)y \in B(x_0, r)$. $\square$

More generally, an ellipsoid of the form

$$E := \{x : (x - x_0)^\top P^{-1}(x - x_0) \leq 1\}, \quad P \succ 0,$$

is convex, since it is the inverse image of a Euclidean ball under the linear map $x \mapsto P^{-1/2}(x - x_0)$.

Example 1.8 (Convex Cones)**.** A set $K \subseteq \mathbb{R}^n$ is called a *cone* if

$$x \in K \;\Rightarrow\; \alpha x \in K \quad \forall \alpha \geq 0.$$

If K is also convex, it is called a *convex cone*.

Proposition 1.9. *If K is a convex cone, then for any $x, y \in K$ and any $\alpha, \beta \geq 0$, we have $\alpha x + \beta y \in K$.*

Proof. Since K is a cone, $\alpha x \in K$ and $\beta y \in K$. Because K is convex, any convex combination of αx and βy lies in K. In particular, for $\alpha + \beta > 0$,

$$\frac{\alpha}{\alpha+\beta}(\alpha+\beta)x + \frac{\beta}{\alpha+\beta}(\alpha+\beta)y = \alpha x + \beta y \in K.$$

The case $\alpha = \beta = 0$ is trivial. $\square$

Typical examples of convex cones include the nonnegative orthant $\mathbb{R}^n_+$, second-order (Lorentz) cones, and the cone of positive semidefinite matrices.

Example 1.10 (Convex Hulls)**.** Given a set $S \subseteq \mathbb{R}^n$, its convex hull is defined as

$$\operatorname{conv}(S) = \left\{ \sum_{i=1}^{k} \lambda_i x_i \;\middle|\; x_i \in S,\ \lambda_i \geq 0,\ \sum_{i=1}^{k} \lambda_i = 1 \right\}.$$

Proposition 1.11. *The convex hull* $\operatorname{conv}(S)$ *is convex and is the smallest convex set containing* S*.*

Proof. Convexity follows from closure under convex combinations: a convex combination of convex combinations is itself a convex combination. Minimality follows from the observation that any convex set containing S must contain all finite convex combinations of points in S, and hence must contain $\operatorname{conv}(S)$. □

Example 1.12 (Descent Cones)**.** For a differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, define the descent set at x by

$$S := \{d \in \mathbb{R}^n : \nabla f(x)^\top d < 0\}.$$

Proposition 1.13. *The descent set* S *is a convex cone.*

Proof. The set S is an open half-space defined by a linear inequality, hence convex. Moreover, if $d \in S$ and $\alpha > 0$, then $\nabla f(x)^\top(\alpha d) = \alpha \nabla f(x)^\top d < 0$, so S is a cone. □

Geometrically, S consists of all directions along which the function decreases to first order at x.

1.1.2 Properties of Convex Sets

Theorem 1.14 (Intersection Property)**.** *If* $\{C_i\}_{i \in \mathcal{P}}$ *is a collection of convex sets, then their intersection*

$$C := \bigcap_{i \in \mathcal{P}} C_i$$

is convex.

Proof. Let $x, y \in C$. Then $x, y \in C_i$ for all $i \in \mathcal{P}$. Since each C_i is convex,

$$\lambda x + (1 - \lambda) y \in C_i \quad \forall i \in \mathcal{P}.$$

Hence $\lambda x + (1 - \lambda) y \in C$. □

Proposition 1.15 (Linear Images and Preimages)**.** *Let* A *be a linear map.*

- *If* C *is convex, then* $A(C)$ *is convex.*

- *If D is convex, then $A^{-1}(D)$ is convex.*

Proof. Both claims follow directly from linearity of A and preservation of convex combinations. □

Convexity is also preserved under Cartesian products.

OTHER PROPERTIES.

1.1.3 Supporting Hyperplanes

Supporting hyperplanes formalize the idea that, at any boundary point of a convex set, one can place a hyperplane that "touches" the set without cutting through it. This result is purely geometric and underlies first-order optimality conditions in convex optimization.

Theorem 1.16 (Supporting Hyperplane Theorem)**.** *Let $C \subset \mathbb{R}^n$ be a nonempty convex set, and let z be a boundary point of C such that $z \notin \operatorname{int}(C)$. Then there exists a nonzero vector $a \in \mathbb{R}^n$ satisfying*

$$a^\top x \leq a^\top z \quad \forall x \in C.$$

Equivalently, the hyperplane

$$H := \{x \in \mathbb{R}^n : a^\top x = a^\top z\}$$

contains the point z and places the entire set C in the closed half-space $\{x : a^\top x \leq a^\top z\}$. Such a hyperplane is called a *supporting hyperplane* of C at z.

Proof. Because z is a boundary point of C, there exists a sequence $\{z_k\}_{k=1}^\infty \subset \mathbb{R}^n \setminus C$ such that $z_k \to z$ as $k \to \infty$. For each k, the singleton set $\{z_k\}$ and the set C are nonempty, convex, and disjoint.

We first note that C can be replaced by its closure $\overline{C}$ without loss of generality: convexity is preserved under closure, and any supporting hyperplane for $\overline{C}$ at z also supports C. Hence we may assume that C is closed.

By the strong separation theorem for a closed convex set and a point not belonging to it (a special case of the Hahn–Banach separation theorem), there exist, for each k, a nonzero vector $a_k \in \mathbb{R}^n$ and a scalar $b_k \in \mathbb{R}$ such that

$$a_k^\top x \leq b_k < a_k^\top z_k \quad \forall x \in C. \tag{1.2}$$

We may normalize each a_k so that $\|a_k\|_2 = 1$. Since the unit sphere in $\mathbb{R}^n$ is compact, there exists a subsequence $\{a_{k_j}\}$ that converges to some vector $a \in \mathbb{R}^n$ with $\|a\|_2 = 1$, and hence $a \neq 0$.

Fix an arbitrary point $x \in C$. From (1.2), for each index j, we have the chain of inequalities

$$a_{k_j}^\top x \leq b_{k_j} < a_{k_j}^\top z_{k_j}.$$

The left-hand inequality implies that the real sequence $\{a_{k_j}^\top x\}_{j=1}^\infty$ is bounded above by b_{k_j} for every j. Since $a_{k_j} \to a$ in $\mathbb{R}^n$, continuity of the inner product yields

$$\lim_{j\to\infty} a_{k_j}^\top x = a^\top x.$$

Applying the lim inf operator to the inequality $a_{k_j}^\top x \leq b_{k_j}$ and using the fact that limits preserve inequalities under lim inf, we obtain

$$a^\top x = \lim_{j\to\infty} a_{k_j}^\top x \leq \liminf_{j\to\infty} b_{k_j}.$$

Similarly, the strict inequality $b_{k_j} < a_{k_j}^\top z_{k_j}$ implies that $\{b_{k_j}\}$ is bounded above by the sequence $\{a_{k_j}^\top z_{k_j}\}$. Since $a_{k_j} \to a$ and $z_{k_j} \to z$, continuity of the inner product again gives

$$\lim_{j\to\infty} a_{k_j}^\top z_{k_j} = a^\top z.$$

Taking limits in the inequality $b_{k_j} < a_{k_j}^\top z_{k_j}$ and using the definition of lim inf, we conclude that

$$\liminf_{j\to\infty} b_{k_j} \leq \lim_{j\to\infty} a_{k_j}^\top z_{k_j} = a^\top z.$$

Combining these inequalities yields

$$a^\top x \leq \liminf_{j\to\infty} b_{k_j} \leq a^\top z.$$

Because $x \in C$ was chosen arbitrarily, this inequality holds for all $x \in C$, and therefore

$$a^\top x \leq a^\top z \quad \forall x \in C.$$

This establishes the existence of a supporting hyperplane of C at the point z with normal vector a.

□

1.1.4 Separating Hyperplanes

Separating hyperplanes describe the relative geometry of two convex sets. They formalize the intuition that two disjoint convex sets can always be *separated* by a hyperplane, meaning that the sets lie on opposite sides of the hyperplane (possibly touching it).

Theorem 1.17 (Separating Hyperplane Theorem)**.** *Let C and D be two nonempty, disjoint convex sets in $\mathbb{R}^n$. Then there exist a nonzero vector $a \in \mathbb{R}^n$ and a scalar $b \in \mathbb{R}$ such that*

$$a^\top x \leq b \quad \forall x \in C, \qquad a^\top x \geq b \quad \forall x \in D.$$

Proof. Define the Minkowski difference

$$E := C - D := \{x - y : x \in C,\ y \in D\}.$$

Step 1: Convexity of E. Let $u_1 = x_1 - y_1$ and $u_2 = x_2 - y_2$ be arbitrary elements of E, with $x_1, x_2 \in C$ and $y_1, y_2 \in D$. For any $\lambda \in [0, 1]$, we have

$$\lambda u_1 + (1 - \lambda)u_2 = (\lambda x_1 + (1 - \lambda)x_2) - (\lambda y_1 + (1 - \lambda)y_2).$$

Since C and D are convex, $\lambda x_1 + (1 - \lambda)x_2 \in C$ and $\lambda y_1 + (1 - \lambda)y_2 \in D$, which implies $\lambda u_1 + (1 - \lambda)u_2 \in E$. Hence E is convex.

Step 2: The origin does not belong to E. If $0 \in E$, then there exist $x \in C$ and $y \in D$ such that $x - y = 0$, i.e., $x = y$. This implies $C \cap D \neq \emptyset$, contradicting the assumption that C and D are disjoint. Therefore,

$$0 \notin E.$$

Step 3: The origin is not an interior point of E. Since $0 \notin E$, it follows immediately that $0 \notin \operatorname{int}(E)$. In particular, there exists no open ball centered at the origin that is contained in E.

Step 4: Supporting hyperplane at the origin. Because E is a convex set and $0 \notin \operatorname{int}(E)$, the Supporting Hyperplane Theorem applies. Hence, there exists a nonzero vector $a \in \mathbb{R}^n$ such that

$$a^\top u \leq 0 \quad \forall u \in E.$$

Step 5: Separation of C and D. For arbitrary $x \in C$ and $y \in D$, the vector $u := x - y$ belongs to E, and thus

$$a^\top (x - y) \leq 0,$$

which implies

$$a^\top x \leq a^\top y \quad \forall x \in C,\ y \in D.$$

Define

$$b := \sup_{x \in C} a^\top x \in [-\infty, +\infty].$$

Then, by definition,

$$a^\top x \leq b \quad \forall x \in C.$$

Moreover, the inequality $a^\top x \leq a^\top y$ for all $x \in C$ and $y \in D$ implies

$$a^\top y \geq b \quad \forall y \in D.$$

Therefore, the hyperplane

$$H := \{x \in \mathbb{R}^n : a^\top x = b\}$$

separates the sets C and D, completing the proof. □

1.2 Convex Functions

Convex functions play a central role in optimization, economics, control, and learning theory. Their importance stems from the fact that convexity imposes a strong global structure: local information (such as gradients or subgradients) is sufficient to characterize global behavior. In particular, convexity rules out pathological phenomena such as spurious local minima and ensures that local optimality conditions are also globally meaningful. This structural simplicity underlies the tractability of a wide class of optimization problems encountered in theory and practice.

Let $C \subseteq \mathbb{R}^n$ be a convex set, and let

$$f : C \to \mathbb{R}$$

be a real-valued function.

Definition 1.18 (Convex Function)**.** The function f is said to be *convex* on C if for all $x, y \in C$

and all $\lambda \in [0,1]$,

$$f(\lambda x + (1-\lambda)y) \;\leq\; \lambda f(x) + (1-\lambda) f(y). \tag{1.3}$$

Inequality (1.3) expresses the fundamental idea that the function value at any convex combination of two points does not exceed the corresponding convex combination of the function values. Equivalently, linear interpolation in the domain upper-bounds interpolation in the range.

Geometrically, (1.3) states that the graph of f lies below the straight line segment connecting $(x, f(x))$ and $(y, f(y))$. Thus, a convex function has a "bowl-shaped" geometry, possibly with flat regions.

To formalize this geometric viewpoint, we introduce the notion of the epigraph.

Definition 1.19 (Epigraph)**.** The *epigraph* of a function $f : C \to \mathbb{R}$ is the set

$$\operatorname{epi}(f) \;:=\; \{(x,t) \in \mathbb{R}^{n+1} \mid x \in C,\; t \geq f(x)\}.$$

The epigraph consists of all points lying on or above the graph of f. It provides a direct way to view functions as sets in a higher-dimensional space.

Theorem 1.20 (Epigraph Characterization of Convexity)**.** *A function $f : C \to \mathbb{R}$ is convex on a convex set C if and only if its epigraph* $\operatorname{epi}(f)$ *is a convex subset of* $\mathbb{R}^{n+1}$.

This characterization establishes a precise equivalence between convex functions and convex sets. It plays a central role in duality theory, constrained optimization, and variational analysis, where optimization problems are often studied through the geometry of epigraphs.

Example 1.21 (Convexity preserved under partial minimization)**.** Let $f(x,y)$ be convex in the joint variables (x,y) on a convex set $C \subseteq \mathbb{R}^n \times \mathbb{R}^m$. Define the marginal function

$$g(x) := \inf_y f(x,y),$$

allowing g to take the value $+\infty$. Then g is convex.

Proof. The epigraph of g is

$$\operatorname{epi}(g) = \{(x,t) \in \mathbb{R}^{n+1} \mid \exists\, y \text{ such that } f(x,y) \leq t\}.$$

Equivalently,

$$\operatorname{epi}(g) = \operatorname{proj}_{(x,t)}(\operatorname{epi}(f)),$$

the projection of the epigraph of f onto the (x, t)-coordinates.

Since f is convex, $\mathrm{epi}(f)$ is convex. Projections preserve convexity, hence $\mathrm{epi}(g)$ is convex. By the epigraph characterization of convexity, g is convex.

Example 1.22 (Jensen's inequality via epigraph geometry)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and let X be an $\mathbb{R}^n$-valued random variable with finite expectation. Then

$$f\big(\mathbb{E}[X]\big) \;\leq\; \mathbb{E}\big[f(X)\big].$$

Proof. For each realization of X, the point $(X, f(X))$ lies in the epigraph $\mathrm{epi}(f)$. Since $\mathrm{epi}(f)$ is convex, it is closed under convex combinations. Taking expectations yields

$$\mathbb{E}(X, f(X)) = \big(\mathbb{E}[X], \mathbb{E}[f(X)]\big) \in \mathrm{epi}(f).$$

By definition of the epigraph, this implies $f\big(\mathbb{E}[X]\big) \leq \mathbb{E}\big[f(X)\big]$, which is Jensen's inequality.

Definition 1.23 (Strictly Convex Function)**.** The function f is *strictly convex* on C if for all $x \neq y$ in C and all $\lambda \in (0, 1)$,

$$f(\lambda x + (1-\lambda)y) \;<\; \lambda f(x) + (1-\lambda) f(y).$$

Strict convexity excludes flat directions: every chord of the graph lies *strictly* above the function except at its endpoints. As a consequence, strictly convex functions admit at most one minimizer on a convex domain, a property that is particularly important in estimation, control, and learning applications where uniqueness of solutions is desirable.

Convexity alone already implies strong regularity. Even in the absence of differentiability, convex functions enjoy continuity and well-behaved local structure on the interior of their domains.

Theorem 1.24 (Continuity on the Interior)**.** *If f is convex on a convex set C, then f is continuous on* $\mathrm{int}(C)$.

Proof. Let $x_0 \in \mathrm{int}(C)$. By definition of interior, there exists $\epsilon > 0$ such that the closed ball

$$B_\epsilon(x_0) \subset C.$$

Fix any $x \in B_{\epsilon/2}(x_0)$. For any direction d with $\|d\| \leq \epsilon/2$, both points $x + d$ and $x - d$

belong to $B_\epsilon(x_0) \subset C$. By convexity,

$$f(x) \;\leq\; \frac{1}{2}f(x+d) + \frac{1}{2}f(x-d).$$

Rearranging yields

$$|f(x+d) - f(x)| \leq f(x+d) + f(x-d) - 2f(x),$$

which shows that oscillations of f near x are controlled by its values on the compact set $B_\epsilon(x_0)$.

Since a convex function is bounded above on compact subsets of the interior, there exists $M > 0$ such that

$$|f(y)| \leq M \qquad \forall y \in B_\epsilon(x_0).$$

Using the convexity inequality with x_0 and $y \in B_{\epsilon/2}(x_0)$, one obtains a local Lipschitz bound:

$$|f(y) - f(x_0)| \leq \frac{2M}{\epsilon}\|y - x_0\|.$$

Thus f is locally Lipschitz in a neighborhood of x_0, and hence continuous at x_0. Since $x_0 \in \mathrm{int}(C)$ was arbitrary, f is continuous on $\mathrm{int}(C)$. □

This result ensures that convex functions cannot exhibit abrupt oscillations or discontinuities in the interior of their domain. In particular, small changes in the argument lead to small changes in function values, which is essential for the stability of optimization algorithms.

Example 1.25 (Failure of continuity at the boundary)**.** Let

$$C := (0,1] \subset \mathbb{R}, \qquad f(x) := \frac{1}{x}.$$

The function f is convex on C, since

$$f''(x) = \frac{2}{x^3} > 0 \qquad \forall x \in (0,1].$$

However, f is not continuous at the boundary point $x = 0$, which does not belong to C. Indeed,

$$\lim_{x \downarrow 0} f(x) = +\infty.$$

Thus, although f is continuous on the interior $\mathrm{int}(C) = (0,1)$, it cannot be extended continu-

ously to the boundary of C.

This example illustrates that convexity alone does not control behavior at boundary points, and that the continuity result in Theorem 1.24 is sharp.

Remark 1.26. Convex functions may fail to be continuous on the boundary of C. However, interior continuity is sufficient for most optimization arguments, including the derivation of optimality conditions and the convergence analysis of iterative methods.

In subsequent sections, we refine the notion of convexity by introducing first-order and second-order characterizations, including gradient inequalities, subgradients, and Hessian conditions. These analytical tools make precise the connection between the geometric definition given above and practical methods for solving convex optimization problems.

1.2.1 First-Order Characterization of Convexity

When the function f is differentiable, convexity admits a particularly powerful and intuitive characterization in terms of first-order information. This characterization reveals that for convex functions, local linear approximations provide global underestimates of the function, thereby linking geometry, analysis, and optimization in a precise way.

Theorem 1.27 (First-Order Condition for Convexity)**.** *Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable. Then f is convex if and only if*

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \textit{for all } x, y \in \mathbb{R}^n. \tag{1.4}$$

This inequality is often called the *gradient inequality*. It asserts that the function always lies above its first-order Taylor expansion at any point. Unlike general smooth functions, for which Taylor expansions are only locally accurate, convex functions enjoy a global linear lower bound at every point.

Inequality (1.4) states that the tangent hyperplane to the graph of f at x lies below the graph of f everywhere. In geometric terms, the graph of a convex function has the property that no tangent plane can cut through it. This global supporting-hyperplane property is the functional analog of supporting hyperplanes for convex sets and forms the basis of many optimality conditions.

Proof. ($\Rightarrow$) Assume f is convex. Fix arbitrary points $x, y \in \mathbb{R}^n$ and let $\lambda \in (0, 1)$. By convexity,

$$f\big(y + \lambda(x - y)\big) \leq (1 - \lambda) f(y) + \lambda f(x).$$

Rewriting this inequality yields

$$f\big(y+\lambda(x-y)\big)-f(y)\le\lambda\big(f(x)-f(y)\big).$$

Dividing both sides by $\lambda>0$ gives

$$\frac{f(y+\lambda(x-y))-f(y)}{\lambda}\le f(x)-f(y).$$

Taking the limit as $\lambda\to 0^+$ and using differentiability of f at y, we obtain

$$\nabla f(y)^\top(x-y)\le f(x)-f(y),$$

which, after rearranging terms, is equivalent to (1.4).

($\Leftarrow$) Assume that (1.4) holds for all $x,y\in\mathbb{R}^n$. Let $x,y\in\mathbb{R}^n$ and let $\lambda\in[0,1]$. Define

$$x':=\lambda x+(1-\lambda)y.$$

Applying (1.4) at the point x' with $y=x$ and $y=y$, respectively, yields

$$\nabla f(x')^\top(x-x')\le f(x)-f(x'),\qquad \nabla f(x')^\top(y-x')\le f(y)-f(x').$$

Multiplying the first inequality by λ and the second by $(1-\lambda)$ and summing, we obtain

$$\nabla f(x')^\top\big(\lambda(x-x')+(1-\lambda)(y-x')\big)\le\lambda f(x)+(1-\lambda)f(y)-f(x').$$

Since $\lambda(x-x')+(1-\lambda)(y-x')=0$, the left-hand side vanishes, and we are left with

$$f(x')\le\lambda f(x)+(1-\lambda)f(y),$$

which is precisely the convexity inequality (1.3). □

1.2.2 Second-Order Characterization

When second derivatives exist, convexity can be verified locally via curvature properties encoded in the Hessian matrix.

Theorem 1.28 (Hessian Test)**.** *Let $f\in C^2(\mathbb{R}^n)$.*

- *f is convex if and only if $\nabla^2 f(x) \succeq 0$ for all $x \in \mathbb{R}^n$;*
- *f is strictly convex if $\nabla^2 f(x) \succ 0$ for all $x \in \mathbb{R}^n$.*

Positive semidefiniteness of the Hessian means that the second directional derivative of f is nonnegative in every direction, so the function never bends downward. Positive definiteness strengthens this condition by enforcing strictly positive curvature in every nonzero direction, thereby ruling out flat regions.

Example 1.29. Consider the scalar function $f(t) = t^4$. Then

$$f''(t) = 12t^2 \geq 0 \quad \forall t,$$

so f is convex. Moreover, $f''(t) > 0$ for all $t \neq 0$, implying that f is strictly convex, even though the Hessian vanishes at $t = 0$.

1.2.3 Equivalent Conditions for Strict Convexity

Assume $f \in C^1(\mathbb{R}^n)$. The following statements are equivalent:

1. f is strictly convex;
2. $$f(y) > f(x) + \nabla f(x)^\top (y - x), \quad \forall x \neq y;$$
3. $$(\nabla f(x) - \nabla f(y))^\top (x - y) > 0, \quad \forall x \neq y.$$

The third condition expresses strict monotonicity of the gradient mapping, a property that plays an important role in variational inequalities and equilibrium theory.

If $f \in C^2(\mathbb{R}^n)$, these conditions are further equivalent to

$$\nabla^2 f(x) \succ 0 \quad \forall x \in \mathbb{R}^n.$$

2 Introduction to Convex Optimization

Convex optimization concerns a special class of optimization problems in which both the objective function and the feasible set possess convexity properties. These properties impose a

strong global structure on the problem and lead to powerful theoretical guarantees that sharply distinguish convex programs from general, nonconvex optimization problems.

Throughout this section, we consider optimization problems of the form

$$\min_{x \in \mathcal{F}} f(x),$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is a convex function and the feasible set $\mathcal{F} \subseteq \mathbb{R}^n$ is defined by convex inequality constraints. We assume that $\mathcal{F}$ is nonempty.

2.1 Convexity of the Feasible Set

Theorem 2.1 (Convexity of the Feasible Set)**.** *Let*

$$\mathcal{F} := \{x \in \mathbb{R}^n : g_i(x) \leq 0,\ i \in \mathcal{Q}\},$$

where each $g_i : \mathbb{R}^n \to \mathbb{R}$ is a convex function. Then the feasible set $\mathcal{F}$ is a convex set.

Proof. Fix any index $i \in \mathcal{Q}$. Since g_i is convex, its sublevel set

$$\{x \in \mathbb{R}^n : g_i(x) \leq 0\}$$

is convex by the definition of convexity in (1.3). Indeed, for any x, y in this set and any $\lambda \in [0, 1]$, convexity implies

$$g_i(\lambda x + (1 - \lambda)y) \leq \lambda g_i(x) + (1 - \lambda)g_i(y) \leq 0.$$

Thus $\lambda x + (1 - \lambda)y$ also belongs to the set.

The feasible set $\mathcal{F}$ is the intersection of these convex sublevel sets over all $i \in \mathcal{Q}$. Since arbitrary intersections of convex sets are convex, it follows that $\mathcal{F}$ is convex. □

Convexity of the feasible set ensures that any two feasible points can be joined by a line segment that remains entirely feasible. This geometric property is crucial for both theoretical analysis and the design of optimization algorithms, as it allows global conclusions to be drawn from local behavior.

2.2 Local Optimality Implies Global Optimality

One of the most striking and useful consequences of convexity is that local optimality automatically implies global optimality.

Theorem 2.2 (Local Minima Are Global)**.** *For a convex optimization problem, every local minimum is also a global minimum.*

Proof. Let $x^\star \in \mathcal{F}$ be a local minimum of f over $\mathcal{F}$, and let $y \in \mathcal{F}$ be arbitrary. Because $\mathcal{F}$ is convex by Theorem 2.1, for any $\lambda \in (0,1)$ the point

$$x_\lambda := \lambda x^\star + (1-\lambda)y$$

also belongs to $\mathcal{F}$.

By convexity of f (cf. (1.3)),

$$f(x_\lambda) \leq \lambda f(x^\star) + (1-\lambda)f(y).$$

Since $x^\star$ is a local minimum, there exists a neighborhood of $x^\star$ within which $f(x) \geq f(x^\star)$ for all feasible x. For λ sufficiently close to 1, the point x_λ lies in this neighborhood, and hence

$$f(x_\lambda) \geq f(x^\star).$$

Combining the two inequalities yields

$$\lambda f(x^\star) + (1-\lambda)f(y) \geq f(x^\star),$$

which simplifies to

$$(1-\lambda)f(y) \geq (1-\lambda)f(x^\star).$$

Since $1-\lambda > 0$, we conclude that $f(y) \geq f(x^\star)$. Because $y \in \mathcal{F}$ was arbitrary, $x^\star$ is a global minimum. □

Remark 2.3. This result is one of the defining advantages of convex optimization: there are no spurious local minima. In contrast, nonconvex optimization problems may exhibit many local minima that are not globally optimal.

Theorem 2.4 (Uniqueness of the Minimizer)**.** *If f is strictly convex, then the global minimizer (if it exists) is unique.*

Proof. Suppose there exist two distinct minimizers $x_1 \neq x_2$ such that

$$f(x_1) = f(x_2) = \inf_x f(x).$$

For any $\lambda \in (0,1)$, strict convexity implies

$$f(\lambda x_1 + (1-\lambda)x_2) < \lambda f(x_1) + (1-\lambda) f(x_2),$$

which contradicts optimality. Hence, the minimizer is unique. $\square$

Remark 2.5. For convex functions,

$$\text{local minimum} \iff \text{global minimum}.$$

This equivalence fails in nonconvex optimization, where local optimality may provide little information about global behavior.

3 Constraint Qualifications and Slater's Condition

To derive first-order and saddle-point optimality conditions such as the Karush–Kuhn–Tucker (KKT) conditions for the convex program (1.1), additional regularity assumptions on the constraint system are required. These assumptions are collectively known as *constraint qualifications*. Among them, Slater's condition plays a fundamental role in convex optimization: it ensures strong duality, guarantees the existence of Lagrange multipliers, and allows global optimality to be characterized via local conditions.

3.1 Slater's Condition

Consider the convex optimization problem (1.1), where f and g_i, $i \in \mathcal{Q}$, are convex functions.

Definition 3.1 (Slater's Condition)**.** The convex program (1.1) is said to satisfy *Slater's condition* if there exists a point $\bar{x} \in \mathbb{R}^n$ such that

$$g_i(\bar{x}) < 0 \quad \text{for all } i \in \mathcal{Q}.$$

Such a point $\bar{x}$ is called a *strictly feasible point*.

Slater's condition requires that all inequality constraints be satisfied strictly at some point. Equivalently, the feasible set

$$\mathcal{F} := \{x \in \mathbb{R}^n : g_i(x) \leq 0,\ i \in \mathcal{Q}\}$$

has a nonempty interior. This interiority condition rules out degenerate constraint geometries in which feasibility is confined entirely to the boundary of the constraint set.

Equality constraints can never satisfy Slater's condition, since they cannot be satisfied strictly. Consequently, Slater's condition is imposed only on inequality constraints. In the presence of equality constraints, strict feasibility is understood relative to the affine subspace defined by those equalities.

3.2 Geometric Characterization

Let $x^\star \in \mathcal{F}$ be a feasible point and define the active constraint set at $x^\star$ by

$$\mathcal{Q}(x^\star) := \{i \in \mathcal{Q} : g_i(x^\star) = 0\}.$$

Theorem 3.2 (Directional Characterization of Slater's Condition)**.** *Assume that each g_i is differentiable. Slater's condition holds if and only if, for every boundary point $x^\star \in \mathcal{F}$, there exists a direction $d \in \mathbb{R}^n$ such that*

$$\nabla g_i(x^\star)^\top d < 0, \quad \forall i \in \mathcal{Q}(x^\star).$$

Proof. Assume first that Slater's condition holds, and let $\bar{x}$ be a strictly feasible point. Fix any boundary point $x^\star \in \mathcal{F}$ and define $d := \bar{x} - x^\star$. For any $i \in \mathcal{Q}(x^\star)$, we have $g_i(x^\star) = 0$ and $g_i(\bar{x}) < 0$. By convexity of g_i,

$$g_i(x^\star + \alpha d) = g_i((1-\alpha)x^\star + \alpha\bar{x}) \leq (1-\alpha)g_i(x^\star) + \alpha g_i(\bar{x}) = \alpha g_i(\bar{x}) < 0$$

for all sufficiently small $\alpha > 0$. Dividing by α and letting $\alpha \to 0^+$ yields

$$\nabla g_i(x^\star)^\top d < 0, \quad \forall i \in \mathcal{Q}(x^\star).$$

Conversely, suppose that the stated directional condition holds at every boundary point. Fix any boundary point $x^\star \in \mathcal{F}$. By assumption, there exists a direction d such that

$$\nabla g_i(x^\star)^\top d < 0 \quad \forall i \in \mathcal{Q}(x^\star).$$

By differentiability and continuity of each g_i, there exists $\epsilon > 0$ such that

$$g_i(x^\star + \alpha d) < 0 \quad \forall i \in \mathcal{Q}(x^\star), \quad \forall \alpha \in (0, \epsilon).$$

For constraints $i \notin \mathcal{Q}(x^\star)$, we already have $g_i(x^\star) < 0$, and by continuity this inequality remains strict for sufficiently small $\alpha > 0$. Hence $x^\star + \alpha d$ is strictly feasible, and Slater's condition holds. □

Geometrically, Theorem 3.2 states that at every boundary point of the feasible region, there exists a direction that points strictly into the interior of $\mathcal{F}$.

Corollary 3.3. *If, for every boundary point $x^\star \in \mathcal{F}$, the set of gradients*

$$\{\nabla g_i(x^\star) : i \in \mathcal{Q}(x^\star)\}$$

is linearly independent, then Slater's condition holds.

Proof. Fix any boundary point $x^\star \in \mathcal{F}$. Linear independence of the gradients implies that there exists no nontrivial vector of coefficients $\lambda_i \geq 0$, not all zero, such that

$$\sum_{i \in \mathcal{Q}(x^\star)} \lambda_i \nabla g_i(x^\star) = 0.$$

By Gordan's theorem of alternatives, this is equivalent to the existence of a vector $d \in \mathbb{R}^n$ satisfying

$$\nabla g_i(x^\star)^\top d < 0 \quad \forall i \in \mathcal{Q}(x^\star).$$

The conclusion then follows directly from Theorem 3.2. □

3.3 Minimal Index Sets and Constraint Reduction

Define the *minimal active index set*

$$\mathcal{Q}^= := \{i \in \mathcal{Q} : g_i(x) = 0 \text{ for all } x \in \mathcal{F}\}.$$

Constraints in $\mathcal{Q}^=$ are always active and cannot be satisfied strictly.

Define the complementary index set

$$\mathcal{Q}^< := \mathcal{Q} \setminus \mathcal{Q}^=.$$

Proposition 3.4. *The feasible set admits the representation*

$$\mathcal{F} = \{x : g_i(x) \leq 0,\ i \in \mathcal{Q}^<\} \cap \{x : g_i(x) = 0,\ i \in \mathcal{Q}^=\}.$$

Proof. By definition of $\mathcal{Q}^{=}$, each constraint indexed by $i \in \mathcal{Q}^{=}$ is active at every feasible point and hence acts as an equality constraint. Conversely, all constraints in $\mathcal{Q}^{<}$ are inactive at least at one feasible point and therefore may admit strict satisfaction. The stated representation follows by partitioning the constraint set accordingly. □

Remark 3.5. Slater's condition holds if and only if $\mathcal{Q}^{=} = \emptyset$. In this sense, Slater's condition excludes degenerate inequality constraints that force feasibility to lie entirely on the boundary of the constraint set. Equality constraints are handled separately and do not obstruct Slater's condition when strict feasibility holds relative to their affine hull.

4 Basic Convex Program Characterization

We now present a fundamental characterization of optimal solutions to convex programs in terms of saddle points of the Lagrangian. This result provides the conceptual bridge between primal optimality, dual variables, and KKT conditions.

Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \qquad i = 1, \ldots, m, \end{aligned} \tag{4.1}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ and $g_i : \mathbb{R}^n \to \mathbb{R}$ are convex functions.

The associated Lagrangian is defined by

$$L(x, \mu) := f(x) + \sum_{i=1}^{m} \mu_i g_i(x), \qquad \mu \in \mathbb{R}^m_+. \tag{4.2}$$

Theorem 4.1 (Saddle-Point Characterization of Optimality)**.** *Assume that Slater's condition holds for (4.1). A point $x^\star \in \mathbb{R}^n$ is a primal optimal solution of (4.1) if and only if there exists a vector of Lagrange multipliers $\mu^\star \in \mathbb{R}^m_+$ such that $(x^\star, \mu^\star)$ is a saddle point of the Lagrangian, i.e.,*

$$L(x^\star, \mu) \;\leq\; L(x^\star, \mu^\star) \;\leq\; L(x, \mu^\star), \quad \forall x \in \mathbb{R}^n,\ \forall \mu \in \mathbb{R}^m_+. \tag{4.3}$$

Proof. We prove necessity and sufficiency.

Necessity. Assume $x^\star$ is optimal for (4.1). Define two sets in $\mathbb{R}^{m+1}$ (with the product order $\geq$ and $\leq$ interpreted componentwise):

$$C_1 := \left\{ y \in \mathbb{R}^{m+1} : \exists x \in \mathbb{R}^n \text{ such that } y \geq \begin{bmatrix} f(x) \\ f_1(x) \\ \vdots \\ f_m(x) \end{bmatrix} \right\}, \qquad C_2 := \left\{ y \in \mathbb{R}^{m+1} : y \leq \begin{bmatrix} f(x^\star) \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right\}.$$

We first verify that C_1 and C_2 are convex. Convexity of C_2 is immediate since it is an intersection of half-spaces of the form $\{y : y_j \leq c_j\}$. For C_1, take any $y^{(1)}, y^{(2)} \in C_1$. Then there exist $x^{(1)}, x^{(2)}$ such that

$$y^{(k)} \geq \begin{bmatrix} f(x^{(k)}) \\ f_1(x^{(k)}) \\ \vdots \\ f_m(x^{(k)}) \end{bmatrix}, \quad k = 1, 2.$$

For any $\lambda \in [0, 1]$, let $x^\lambda := \lambda x^{(1)} + (1 - \lambda) x^{(2)}$. By convexity of f and f_i,

$$\lambda \begin{bmatrix} f(x^{(1)}) \\ f_1(x^{(1)}) \\ \vdots \\ f_m(x^{(1)}) \end{bmatrix} + (1 - \lambda) \begin{bmatrix} f(x^{(2)}) \\ f_1(x^{(2)}) \\ \vdots \\ f_m(x^{(2)}) \end{bmatrix} \geq \begin{bmatrix} f(x^\lambda) \\ f_1(x^\lambda) \\ \vdots \\ f_m(x^\lambda) \end{bmatrix}.$$

Since also $\lambda y^{(1)} + (1 - \lambda) y^{(2)}$ dominates the left-hand side componentwise, we obtain

$$\lambda y^{(1)} + (1 - \lambda) y^{(2)} \geq \begin{bmatrix} f(x^\lambda) \\ f_1(x^\lambda) \\ \vdots \\ f_m(x^\lambda) \end{bmatrix},$$

which shows $\lambda y^{(1)} + (1 - \lambda) y^{(2)} \in C_1$. Hence C_1 is convex.

Next we show that C_1 and C_2 are disjoint. If $y \in C_1 \cap C_2$, then for some x,

$$f(x) \leq f(x^\star), \qquad f_i(x) \leq 0 \; \forall i.$$

Thus x is feasible and achieves objective value no larger than $x^\star$. By optimality of $x^\star$, this

forces $f(x) = f(x^\star)$, so x is also optimal. In particular, C_1 and C_2 have no point that witnesses strict improvement in the first coordinate while keeping the remaining coordinates nonpositive; hence they are disjoint in the sense required for separation.[1]

By the separating hyperplane theorem, there exist $(a_0, a) \neq 0$ and α such that

$$a_0 y_0 + a^\top y \geq \alpha \quad \forall (y_0, y) \in C_1, \qquad a_0 y_0 + a^\top y \leq \alpha \quad \forall (y_0, y) \in C_2.$$

Evaluating the second inequality at the maximal point of C_2 gives

$$a_0 f(x^\star) \leq \alpha,$$

and evaluating the first inequality at $\big(f(x), f_1(x), \ldots, f_m(x)\big)$ yields

$$a_0 f(x) + \sum_{i=1}^{m} a_i f_i(x) \geq \alpha, \quad \forall x \in \mathbb{R}^n.$$

Combining these inequalities gives

$$a_0 f(x) + \sum_{i=1}^{m} a_i f_i(x) \geq a_0 f(x^\star), \quad \forall x \in \mathbb{R}^n. \tag{4.4}$$

We now show $a_0 > 0$ and $a_i \geq 0$. First, $a_0 \neq 0$: if $a_0 = 0$, then (4.4) becomes $\sum_i a_i f_i(x) \geq 0$ for all x. Taking $x = \bar{x}$ from Slater's condition, where $f_i(\bar{x}) < 0$ for all i, forces $\sum_i a_i f_i(\bar{x}) \geq 0$, which is impossible unless $a_i = 0$ for all i, contradicting $(a_0, a) \neq 0$. Hence $a_0 \neq 0$.

Second, $a_i \geq 0$: the set C_1 is upward closed componentwise, meaning that if $y \in C_1$ and $\tilde{y} \geq y$, then $\tilde{y} \in C_1$. If some $a_i < 0$, then for a fixed x, the point

$$y(t) := \begin{bmatrix} f(x) \\ f_1(x) \\ \vdots \\ f_i(x) + t \\ \vdots \\ f_m(x) \end{bmatrix} \in C_1 \quad \forall t \geq 0,$$

but the separating inequality would yield $a_0 y_0(t) + a^\top y(t) \geq \alpha$ for all $t \geq 0$, while the left-hand side tends to $-\infty$ as $t \to \infty$ since $a_i < 0$, a contradiction. Thus $a_i \geq 0$.

[1] A standard way to ensure strict disjointness is to replace C_2 by $\{y : y \leq (f(x^\star) - \epsilon, 0, \ldots, 0)\}$ for $\epsilon > 0$ and then let $\epsilon \downarrow 0$. Slater's condition guarantees the strict separation needed for this limiting argument.

Finally, by possibly multiplying (a_0, a) and α by -1, we may assume $a_0 > 0$. Define $\mu_i^\star := a_i/a_0 \geq 0$. Dividing (4.4) by a_0 gives

$$f(x^\star) \leq f(x) + \sum_{i=1}^{m} \mu_i^\star f_i(x) = L(x, \mu^\star), \quad \forall x,$$

which implies the right inequality in (4.3). The left inequality in (4.3) follows from feasibility of $x^\star$ and $\mu \geq 0$:

$$L(x^\star, \mu) = f(x^\star) + \sum_{i=1}^{m} \mu_i f_i(x^\star) \leq f(x^\star) \leq L(x^\star, \mu^\star).$$

Sufficiency. Conversely, assume there exists $\mu^\star \geq 0$ such that (4.3) holds. Let x be any feasible point. Then $f_i(x) \leq 0$ and $\mu_i^\star \geq 0$ imply $L(x, \mu^\star) \leq f(x)$. Using (4.3),

$$f(x^\star) \leq L(x^\star, \mu^\star) \leq L(x, \mu^\star) \leq f(x),$$

so $x^\star$ attains the minimum of f over the feasible set. Hence $x^\star$ is optimal. □

Remark 4.2. Note that Theorem 4.1 does not assume differentiability of f or f_i. Only convexity and Slater's condition are used, together with the separating hyperplane theorem in $\mathbb{R}^{m+1}$. Differentiability becomes relevant later when one derives first-order KKT conditions in terms of gradients.

Corollary 4.3 (Complementarity Conditions)**.** *Assume Slater's condition holds for the convex program* (4.1)*. If* $(x^\star, \mu^\star)$ *is a saddle point of the Lagrangian, i.e.,* (4.3) *holds, then the following* complementarity conditions *are satisfied:*

$$\mu_i^\star g_i(x^\star) = 0, \qquad i = 1, \ldots, m. \tag{4.5}$$

Proof. Since $x^\star$ is feasible, we have $g_i(x^\star) \leq 0$ for all i, and since $\mu^\star \geq 0$, each product $\mu_i^\star g_i(x^\star)$ is nonpositive.

From the saddle-point inequality (4.3), the left inequality gives

$$L(x^\star, \mu) \leq L(x^\star, \mu^\star) \quad \forall \mu \geq 0.$$

Fix any index i and consider $\mu = \mu^\star + te_i$, where e_i is the ith standard basis vector and $t \geq 0$.

Substituting into the inequality yields

$$f(x^\star) + \sum_{j=1}^{m} \mu_j^\star g_j(x^\star) + t g_i(x^\star) \le f(x^\star) + \sum_{j=1}^{m} \mu_j^\star g_j(x^\star), \quad \forall t \ge 0.$$

This inequality can hold for all $t \ge 0$ only if $g_i(x^\star) = 0$ whenever $\mu_i^\star > 0$. Equivalently,

$$\mu_i^\star g_i(x^\star) = 0 \quad \forall i,$$

which establishes the complementarity conditions. □

Remark 4.4. The complementarity conditions (4.5) express a precise duality balance: each constraint is either *inactive* at the optimum ($g_i(x^\star) < 0$) with zero multiplier, or *active* ($g_i(x^\star) = 0$) with possibly nonzero multiplier. Importantly, this result follows purely from convexity and the saddle-point property and does not require differentiability of the objective or constraint functions.

Theorem 4.5 (Karush–Kuhn–Tucker (KKT) Conditions)**.** *Assume that the objective function f and the constraint functions f_i, $i = 1, \dots, m$, are continuously differentiable. Consider the convex optimization problem (4.1), and suppose that Slater's condition holds. Then a feasible point $x^\star \in \mathbb{R}^n$ is optimal if and only if there exists a vector of Lagrange multipliers $\mu^\star = (\mu_1^\star, \dots, \mu_m^\star) \in \mathbb{R}_+^m$ such that*

$$\text{(Stationarity)} \quad \nabla f(x^\star) + \sum_{i=1}^{m} \mu_i^\star \nabla f_i(x^\star) = 0, \tag{4.6}$$

$$\text{(Primal feasibility)} \quad f_i(x^\star) \le 0, \quad i = 1, \dots, m, \tag{4.7}$$

$$\text{(Dual feasibility)} \quad \mu_i^\star \ge 0, \quad i = 1, \dots, m, \tag{4.8}$$

$$\text{(Complementary slackness)} \quad \mu_i^\star f_i(x^\star) = 0, \quad i = 1, \dots, m. \tag{4.9}$$

Proof. We prove necessity and sufficiency.

Necessity. Assume $x^\star$ is optimal for (4.1). Since Slater's condition holds, Theorem **??** (saddle-point characterization) implies that there exists $\mu^\star \in \mathbb{R}_+^m$ such that

$$L(x^\star, \mu) \le L(x^\star, \mu^\star) \le L(x, \mu^\star), \quad \forall x \in \mathbb{R}^n, \ \forall \mu \in \mathbb{R}_+^m, \tag{4.10}$$

where $L(x, \mu) = f(x) + \sum_{i=1}^m \mu_i f_i(x)$.

Primal feasibility (4.7) holds by assumption ($x^\star$ is feasible). Dual feasibility (4.8) holds by construction ($\mu^\star \geq 0$).

Complementary slackness (4.9) follows from (4.10) as in Corollary 4.3: for each i, take $\mu = \mu^\star + te_i$ with $t \geq 0$ in the left inequality $L(x^\star, \mu) \leq L(x^\star, \mu^\star)$ to obtain

$$f(x^\star) + \sum_{j=1}^{m} \mu_j^\star f_j(x^\star) + t f_i(x^\star) \leq f(x^\star) + \sum_{j=1}^{m} \mu_j^\star f_j(x^\star), \quad \forall t \geq 0,$$

hence $f_i(x^\star) \leq 0$ and $\mu_i^\star > 0 \Rightarrow f_i(x^\star) = 0$, which is equivalent to $\mu_i^\star f_i(x^\star) = 0$.

It remains to prove stationarity (4.6). From the right inequality in (4.10), we have

$$L(x^\star, \mu^\star) \leq L(x, \mu^\star), \quad \forall x \in \mathbb{R}^n,$$

so $x^\star$ is a *global minimizer* of the differentiable function $x \mapsto L(x, \mu^\star)$ over $\mathbb{R}^n$.

We now show that any global minimizer of a differentiable function on $\mathbb{R}^n$ must satisfy zero gradient. Consider the directional function

$$\varphi(t) := L(x^\star + td, \mu^\star), \quad t \in \mathbb{R},$$

for an arbitrary direction $d \in \mathbb{R}^n$. Since $x^\star$ minimizes $L(\cdot, \mu^\star)$ globally, $t = 0$ minimizes φ over $\mathbb{R}$, hence $\varphi'(0) = 0$. By the chain rule,

$$\varphi'(0) = \nabla_x L(x^\star, \mu^\star)^\top d.$$

Since d is arbitrary, this implies $\nabla_x L(x^\star, \mu^\star) = 0$, i.e.,

$$\nabla f(x^\star) + \sum_{i=1}^{m} \mu_i^\star \nabla f_i(x^\star) = 0,$$

which is (4.6). This completes the necessity proof.

Sufficiency. Conversely, assume that there exist $(x^\star, \mu^\star)$ satisfying (4.6)–(4.9). We show that $x^\star$ is optimal.

Fix any $x \in \mathbb{R}^n$. By convexity of f and differentiability, the first-order condition for convexity implies

$$f(x) \geq f(x^\star) + \nabla f(x^\star)^\top (x - x^\star).$$

Similarly, for each i,

$$f_i(x) \geq f_i(x^\star) + \nabla f_i(x^\star)^\top (x - x^\star).$$

Multiply the latter inequalities by $\mu_i^\star \geq 0$ and sum over i:

$$\sum_{i=1}^m \mu_i^\star f_i(x) \geq \sum_{i=1}^m \mu_i^\star f_i(x^\star) + \left(\sum_{i=1}^m \mu_i^\star \nabla f_i(x^\star) \right)^\top (x - x^\star).$$

Adding the inequality for $f(x)$ yields

$$\begin{aligned} L(x, \mu^\star) &= f(x) + \sum_{i=1}^m \mu_i^\star f_i(x) \\ &\geq f(x^\star) + \sum_{i=1}^m \mu_i^\star f_i(x^\star) + \left(\nabla f(x^\star) + \sum_{i=1}^m \mu_i^\star \nabla f_i(x^\star) \right)^\top (x - x^\star). \end{aligned}$$

By stationarity (4.6), the last inner-product term is zero, so

$$L(x, \mu^\star) \geq L(x^\star, \mu^\star), \quad \forall x \in \mathbb{R}^n.$$

This proves the right saddle inequality.

To prove the left saddle inequality, fix any $\mu \geq 0$. Since $x^\star$ is feasible, $f_i(x^\star) \leq 0$, we have

$$L(x^\star, \mu) = f(x^\star) + \sum_{i=1}^m \mu_i f_i(x^\star) \leq f(x^\star).$$

On the other hand, complementary slackness (4.9) implies $\sum_i \mu_i^\star f_i(x^\star) = 0$, hence

$$L(x^\star, \mu^\star) = f(x^\star) + \sum_{i=1}^m \mu_i^\star f_i(x^\star) = f(x^\star).$$

Therefore, for all $\mu \geq 0$,

$$L(x^\star, \mu) \leq L(x^\star, \mu^\star).$$

Combining the two parts gives

$$L(x^\star, \mu) \leq L(x^\star, \mu^\star) \leq L(x, \mu^\star), \quad \forall x \in \mathbb{R}^n, \ \forall \mu \geq 0,$$

so $(x^\star, \mu^\star)$ is a saddle point of the Lagrangian. By the saddle-point characterization (Theorem 4.1), $x^\star$ is optimal for (4.1).

This completes the proof. □

Remark 4.6. The saddle-point characterization (Theorem 4.1) and complementary slackness can be derived without differentiability, using only convexity, Slater's condition, and separation. Differentiability is invoked in Theorem 4.5 only to express stationarity in the gradient form (4.6) and to use the first-order characterization of convex functions in the sufficiency direction.

Example 4.7 (Quadratic Program with a Single Inequality Constraint)**.** Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & \frac{1}{2}(x_1^2 + x_2^2 + x_3^2) \\ \text{subject to} \quad & x_1 + x_2 + x_3 \leq -3. \end{aligned} \tag{4.11}$$

The objective function is strictly convex, with Hessian equal to the identity matrix. The constraint function

$$g(x) := x_1 + x_2 + x_3 + 3$$

is affine and hence convex. The point $\bar{x} = (-2, -1, -1)$ satisfies $g(\bar{x}) < 0$, so Slater's condition holds.

Since the problem is convex and Slater's condition holds, optimality is fully characterized by the Karush–Kuhn–Tucker conditions (4.6)–(4.9).

The Lagrangian is

$$L(x, \mu) = \frac{1}{2}(x_1^2 + x_2^2 + x_3^2) + \mu(x_1 + x_2 + x_3 + 3), \qquad \mu \geq 0.$$

Stationarity (4.6) yields

$$\nabla_x L(x^\star, \mu^\star) = \begin{bmatrix} x_1^\star \\ x_2^\star \\ x_3^\star \end{bmatrix} + \mu^\star \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0,$$

which implies

$$x_1^\star = x_2^\star = x_3^\star = -\mu^\star.$$

Primal feasibility (4.7) requires

$$x_1^\star + x_2^\star + x_3^\star \leq -3.$$

Dual feasibility (4.8) requires $\mu^\star \geq 0$.

Complementary slackness (4.9) gives

$$\mu^\star(x_1^\star + x_2^\star + x_3^\star + 3) = 0.$$

Substituting the stationarity condition into the constraint yields

$$-3\mu^\star \leq -3,$$

so $\mu^\star \geq 1$. Complementary slackness forces the constraint to be active, hence

$$x_1^\star + x_2^\star + x_3^\star = -3.$$

Using $x_i^\star = -\mu^\star$, we obtain $-3\mu^\star = -3$, and therefore

$$\mu^\star = 1, \qquad x_1^\star = x_2^\star = x_3^\star = -1.$$

The unique optimal solution of (4.11) is

$$x^\star = (-1, -1, -1), \qquad \mu^\star = 1.$$

Uniqueness follows from strict convexity of the objective function.

Example 4.8 (Exponential Program with Inequality Constraints)**.** Consider the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & e^{x_1 - x_2} \\ \text{subject to} \quad & e^{x_1} + e^{x_2} \leq 20, \\ & x_1 \geq 0. \end{aligned} \tag{4.12}$$

The objective function $f(x) = e^{x_1 - x_2}$ is convex, since it is the exponential of an affine function. The constraint functions

$$g_1(x) := e^{x_1} + e^{x_2} - 20, \qquad g_2(x) := -x_1$$

are convex. The point $\bar{x} = (0, 0)$ satisfies

$$g_1(\bar{x}) = 2 - 20 < 0, \qquad g_2(\bar{x}) = 0,$$

so Slater's condition holds.

Since the problem is convex and Slater's condition holds, the KKT conditions are necessary and sufficient for optimality.

The Lagrangian is

$$L(x, \mu) = e^{x_1 - x_2} + \mu_1(e^{x_1} + e^{x_2} - 20) + \mu_2(-x_1), \qquad \mu_1, \mu_2 \geq 0.$$

Stationarity yields

$$\nabla_x L(x^\star, \mu^\star) = \begin{bmatrix} e^{x_1^\star - x_2^\star} + \mu_1^\star e^{x_1^\star} - \mu_2^\star \\ -e^{x_1^\star - x_2^\star} + \mu_1^\star e^{x_2^\star} \end{bmatrix} = 0.$$

Primal feasibility requires

$$e^{x_1^\star} + e^{x_2^\star} \leq 20, \qquad x_1^\star \geq 0.$$

Dual feasibility requires $\mu_1^\star, \mu_2^\star \geq 0$.

Complementary slackness gives

$$\mu_1^\star(e^{x_1^\star} + e^{x_2^\star} - 20) = 0, \qquad \mu_2^\star x_1^\star = 0.$$

If both constraints are inactive, then $\mu_1^\star = \mu_2^\star = 0$, which would imply $e^{x_1^\star - x_2^\star} = 0$, a contradiction. If only the exponential constraint is active, stationarity again leads to a contradiction. Hence both constraints must be active, yielding

$$x_1^\star = 0, \qquad e^{x_1^\star} + e^{x_2^\star} = 20.$$

Therefore

$$x_2^\star = \ln 19.$$

The optimal solution of (4.12) is

$$x^\star = (0, \ln 19).$$

Example 4.9 (Quadratic Program with Two Convex Inequality Constraints)**.** Consider the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & x_1^2 + x_2^2 - 4x_1 - 4x_2 \\ \text{subject to} \quad & x_1^2 - x_2 \leq 0, \\ & x_1 + x_2 \leq 2. \end{aligned} \tag{4.13}$$

The objective function is strictly convex, with Hessian equal to $2I$. The constraint functions

$$g_1(x) := x_1^2 - x_2, \qquad g_2(x) := x_1 + x_2 - 2$$

are convex. Slater's condition holds since there exist feasible points with strict inequalities.

The Lagrangian is

$$L(x, \mu) = x_1^2 + x_2^2 - 4x_1 - 4x_2 + \mu_1(x_1^2 - x_2) + \mu_2(x_1 + x_2 - 2), \qquad \mu_1, \mu_2 \geq 0.$$

Stationarity yields

$$\nabla_x L(x^\star, \mu^\star) = \begin{bmatrix} 2x_1^\star - 4 + 2\mu_1^\star x_1^\star + \mu_2^\star \\ 2x_2^\star - 4 - \mu_1^\star + \mu_2^\star \end{bmatrix} = 0.$$

Primal feasibility requires

$$x_1^{\star 2} - x_2^\star \leq 0, \qquad x_1^\star + x_2^\star \leq 2.$$

Dual feasibility requires $\mu_1^\star, \mu_2^\star \geq 0$.

Complementary slackness gives

$$\mu_1^\star(x_1^{\star 2} - x_2^\star) = 0, \qquad \mu_2^\star(x_1^\star + x_2^\star - 2) = 0.$$

If neither constraint is active, then $x^\star = (2, 2)$, which is infeasible. Activating only the nonlinear constraint leads to a solution violating $x_1 + x_2 \leq 2$. Activating the linear constraint with $\mu_1^\star = 0$ yields

$$x_1^\star = x_2^\star = 1, \qquad \mu_2^\star = 2,$$

which satisfies all KKT conditions. The case where both constraints are active reduces to the same solution.

The unique optimal solution of (4.13) is

$$x^\star = (1, 1), \qquad \mu_2^\star = 2.$$

Uniqueness follows from strict convexity of the objective function.

4.1 Sufficiency of KKT Conditions

We illustrate the role of Slater's condition by means of a simple convex optimization problem for which an optimal solution exists, but the Karush–Kuhn–Tucker (KKT) conditions fail to be necessary.

Consider the problem

$$\min_{x \in \mathbb{R}} f(x) = x \quad \text{subject to} \quad f_1(x) = x^2 \leq 0.$$

Feasible set and optimal solution. The constraint $x^2 \leq 0$ implies $x = 0$. Hence the feasible set is

$$\mathcal{F} = \{0\},$$

and the unique feasible point is

$$x^\star = 0,$$

which is therefore the optimal solution.

Failure of Slater's condition. There exists no strictly feasible point, since the constraint $x^2 \leq 0$ is satisfied only at equality for all feasible x. Consequently, Slater's condition does not hold.

Failure of the KKT conditions. The Lagrangian associated with the problem is

$$L(x, \mu) = x + \mu x^2, \qquad \mu \geq 0.$$

The stationarity condition requires

$$\nabla_x L(x, \mu) = 1 + 2\mu x = 0.$$

Evaluating this condition at the optimal point $x^\star = 0$ yields

$$\nabla_x L(0, \mu) = 1 \neq 0 \quad \text{for all } \mu \geq 0.$$

Thus, there exists no Lagrange multiplier $\mu^\star \geq 0$ satisfying the stationarity condition at the optimal solution. The KKT conditions therefore fail to hold at $x^\star$, despite the fact that $x^\star$ is globally optimal.

We now formalize the sufficiency of the KKT conditions for convex optimization problems.

Importantly, no constraint qualification is required for this result.

Theorem 4.10 (Sufficiency of KKT Conditions)**.** *Consider the convex optimization problem*

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \textit{subject to} \quad & f_i(x) \leq 0, \quad i \in \mathcal{P}, \end{aligned} \tag{4.14}$$

where f and f_i, $i \in \mathcal{P}$, are convex and differentiable.

If there exist a feasible point $x^\star$ and multipliers $\mu^\star \geq 0$ such that the KKT conditions hold at $x^\star$, then $x^\star$ is a global minimizer of (4.14).

Proof. The stationarity condition of the KKT system yields

$$\nabla f(x^\star) + \sum_{i \in \mathcal{P}(x^\star)} \mu_i^\star \nabla f_i(x^\star) = 0, \tag{4.15}$$

where $\mathcal{P}(x^\star) := \{i \in \mathcal{P} : f_i(x^\star) = 0\}$ denotes the set of active constraints at $x^\star$.

Let x be any feasible point. Taking the inner product of (4.15) with $(x - x^\star)$ gives

$$0 = \nabla f(x^\star)^\top (x - x^\star) + \sum_{i \in \mathcal{P}(x^\star)} \mu_i^\star \nabla f_i(x^\star)^\top (x - x^\star).$$

By convexity of f and f_i, the first-order inequality implies

$$\nabla f(x^\star)^\top (x - x^\star) \leq f(x) - f(x^\star),$$

and, for each $i \in \mathcal{P}(x^\star)$,

$$\nabla f_i(x^\star)^\top (x - x^\star) \leq f_i(x) - f_i(x^\star).$$

Substituting these inequalities yields

$$0 \leq f(x) - f(x^\star) + \sum_{i \in \mathcal{P}(x^\star)} \mu_i^\star \big(f_i(x) - f_i(x^\star)\big).$$

Since $f_i(x^\star) = 0$ for all active constraints and $f_i(x) \leq 0$ for all feasible x, each term in the summation is nonpositive. Consequently,

$$0 \leq f(x) - f(x^\star),$$

which implies

$$f(x^\star) \le f(x) \quad \text{for all feasible } x.$$

Therefore, $x^\star$ is a global optimal solution of (4.14). □

4.2 Case Study: Water-Filling and the KKT Conditions

We illustrate the Karush–Kuhn–Tucker (KKT) conditions (Theorem 4.5, conditions (4.6)–(4.9)) through the classical *water-filling problem*, which arises in power allocation for parallel channels and in resource allocation problems with a single coupling constraint.

Problem formulation. Let $a_i > 0$, $i = 1, \dots, n$, be given constants. Consider

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & \sum_{i=1}^{n} \log(x_i + a_i) \\ \text{subject to} \quad & \sum_{i=1}^{n} x_i \le P, \\ & x_i \ge 0, \quad i = 1, \dots, n, \end{aligned} \tag{4.16}$$

where $P > 0$ denotes a fixed total resource budget.

The objective function is strictly concave on $\mathbb{R}^n_+$, since each $\log(x_i + a_i)$ is strictly concave and increasing. The feasible set is convex because it is defined by linear inequalities. Hence, problem (4.16) is a convex optimization problem (equivalently, a concave maximization problem).

Slater's condition. Slater's condition holds. Indeed, the point

$$x_i = \frac{P}{2n}, \quad i = 1, \dots, n,$$

satisfies

$$\sum_{i=1}^{n} x_i = \frac{P}{2} < P, \qquad x_i > 0 \quad \forall i.$$

Thus all inequality constraints are satisfied strictly. By Slater's theorem, the KKT conditions are both necessary and sufficient for optimality.

Introduce a multiplier $\mu \ge 0$ associated with the total resource constraint and multipliers

$\nu_i \geq 0$, $i = 1, \ldots, n$, associated with the nonnegativity constraints. The Lagrangian is

$$L(x, \mu, \nu) = \sum_{i=1}^{n} \log(x_i + a_i) - \mu \left(\sum_{i=1}^{n} x_i - P \right) - \sum_{i=1}^{n} \nu_i x_i.$$

KKT conditions. An optimal primal–dual triple $(x^\star, \mu^\star, \nu^\star)$ must satisfy:

Primal feasibility:

$$\sum_{i=1}^{n} x_i^\star \leq P, \qquad x_i^\star \geq 0.$$

Dual feasibility:

$$\mu^\star \geq 0, \qquad \nu_i^\star \geq 0.$$

Stationarity:

$$\frac{1}{x_i^\star + a_i} - \mu^\star - \nu_i^\star = 0, \quad i = 1, \ldots, n. \tag{4.17}$$

Complementary slackness:

$$\nu_i^\star x_i^\star = 0, \quad i = 1, \ldots, n, \qquad \mu^\star \left(\sum_{i=1}^{n} x_i^\star - P \right) = 0. \tag{4.18}$$

We first show that the constraint $\sum_{i=1}^{n} x_i \leq P$ must be active at optimality. Since each $\log(x_i + a_i)$ is strictly increasing in x_i, any feasible point with $\sum_{i=1}^{n} x_i < P$ can be strictly improved by increasing some x_i while remaining feasible. Therefore,

$$\sum_{i=1}^{n} x_i^\star = P,$$

and hence $\mu^\star > 0$ by complementary slackness.

Case analysis for each component. Fix an index $i \in \{1, \ldots, n\}$.

Case 1: $x_i^\star > 0$. Then complementary slackness implies $\nu_i^\star = 0$. Substituting into (4.17) yields

$$\frac{1}{x_i^\star + a_i} = \mu^\star,$$

or equivalently

$$x_i^\star = \frac{1}{\mu^\star} - a_i.$$

Since $x_i^\star > 0$, this expression is valid only if

$$\frac{1}{\mu^\star} > a_i.$$

Case 2: $x_i^\star = 0$. Then stationarity (4.17) implies

$$\nu_i^\star = \frac{1}{a_i} - \mu^\star.$$

Dual feasibility $\nu_i^\star \geq 0$ requires

$$\mu^\star \leq \frac{1}{a_i}, \quad \text{or equivalently} \quad \frac{1}{\mu^\star} \geq a_i.$$

Thus, whenever $\frac{1}{\mu^\star} \leq a_i$, the optimal allocation must satisfy $x_i^\star = 0$.

Combining the two cases yields

$$x_i^\star = \max\left\{0, \ \frac{1}{\mu^\star} - a_i\right\}, \quad i = 1, \ldots, n. \tag{4.19}$$

Substituting (4.19) into the active resource constraint gives the scalar equation

$$\sum_{i=1}^{n} \max\left\{0, \ \frac{1}{\mu^\star} - a_i\right\} = P.$$

The left-hand side is a continuous, strictly decreasing function of $\mu^\star > 0$, with limit $+\infty$ as $\mu^\star \downarrow 0$ and limit 0 as $\mu^\star \to \infty$. Therefore, a unique solution $\mu^\star$ exists.

Each constant a_i represents a baseline height, and $1/\mu^\star$ represents a common water level. Resources are allocated so that all active components reach the same level $1/\mu^\star$, while components with baseline a_i exceeding the water level receive no allocation. The active set is determined precisely by the complementary slackness conditions.

Theorem 4.11 (Optimality and Uniqueness of the Water-Filling Solution)**.** *Consider the water-filling problem (4.16). Let $x^\star$ be defined by (4.19), where the scalar $\mu^\star > 0$ is chosen so that the primal feasibility condition $\sum_{i=1}^n x_i^\star = P$ holds. Then $x^\star$ is the unique global optimal solution of problem (4.16).*

5 Duality and the Dual Function

Duality provides an alternative perspective on constrained optimization problems by associating them with an unconstrained maximization problem in dual variables. This perspective is fundamental for understanding optimality conditions, sensitivity analysis, and algorithmic methods.

Throughout this section, consider the convex optimization problem (1.1) and its associated

Lagrangian

$$\mathcal{L}(x,\mu) = f(x) + \sum_{i\in\mathcal{Q}} \mu_i g_i(x), \quad \mu \geq 0,$$

where $\mu \in \mathbb{R}_+^{|\mathcal{Q}|}$ denotes the vector of Lagrange multipliers.

5.1 The Dual Function and Dual Problem

Definition 5.1 (Dual Function)**.** The *dual function* associated with the convex program (1.1) is defined by

$$d(\mu) := \inf_{x\in\mathbb{R}^n} \mathcal{L}(x,\mu), \qquad \mu \geq 0, \tag{5.1}$$

where $\mathcal{L}(x,\mu)$ denotes the Lagrangian of the primal problem.

For each fixed $\mu \geq 0$, the value $d(\mu)$ provides a lower bound on the optimal value of the primal problem, i.e.,

$$d(\mu) \leq f^\star,$$

where $f^\star$ denotes the optimal primal value. This property is known as *weak duality*.

Definition 5.2 (Dual Problem)**.** The *dual problem* corresponding to (1.1) is defined as

$$\sup_{\mu\geq 0} \; d(\mu). \tag{5.2}$$

The optimal value of the dual problem is denoted by

$$d^\star := \sup_{\mu\geq 0} d(\mu).$$

5.2 Properties of Dual Functions

Theorem 5.3 (Weak Duality)**.** *For any dual feasible point $\mu \geq 0$, the dual function satisfies*

$$d(\mu) \leq f^\star,$$

where $f^\star$ is the optimal value of the primal problem (1.1).

Proof. Let $\mu \geq 0$ and let $x^\star$ be any primal feasible point, i.e., $g_i(x^\star) \leq 0$ for all $i \in \mathcal{Q}$. Then

$$\mathcal{L}(x^\star,\mu) = f(x^\star) + \sum_{i\in\mathcal{Q}} \mu_i g_i(x^\star) \leq f(x^\star),$$

since $\mu_i \geq 0$ and $g_i(x^\star) \leq 0$. Taking the infimum over x yields

$$d(\mu) = \inf_x \mathcal{L}(x, \mu) \leq \mathcal{L}(x^\star, \mu) \leq f(x^\star).$$

Since this holds for any feasible $x^\star$, it follows that $d(\mu) \leq f^\star$. □

A key structural property of the dual function is concavity, which holds independently of whether the primal problem is convex.

Theorem 5.4 (Concavity of the Dual Function)**.** *The dual function $d(\mu)$ is concave on $\mathbb{R}_+^{|\mathcal{Q}|}$.*

Proof. Let $\mu^1, \mu^2 \geq 0$ and let $\lambda \in [0, 1]$. Define

$$\mu := \lambda\mu^1 + (1 - \lambda)\mu^2.$$

By definition of the dual function,

$$\begin{aligned} d(\mu) &= \inf_{x \in \mathbb{R}^n} \left(f(x) + \sum_{i \in \mathcal{Q}} \mu_i g_i(x) \right) \\ &= \inf_{x \in \mathbb{R}^n} \left(\lambda \mathcal{L}(x, \mu^1) + (1 - \lambda)\mathcal{L}(x, \mu^2) \right). \end{aligned}$$

Since the infimum of a convex combination of functions is greater than or equal to the same convex combination of their infima, we obtain

$$d(\mu) \geq \lambda \inf_x \mathcal{L}(x, \mu^1) + (1 - \lambda) \inf_x \mathcal{L}(x, \mu^2) = \lambda d(\mu^1) + (1 - \lambda) d(\mu^2).$$

Therefore, $d(\mu)$ is concave. □

Remark 5.5. Although the primal problem (1.1) is a minimization problem, the dual problem (5.2) is always a concave maximization problem, even when the primal problem is nonconvex. This structural asymmetry underlies the power of dual methods in optimization.

Example 5.6 (Quadratic Example Revisited)**.** We revisit the quadratic program introduced earlier in order to illustrate the construction of the dual function and the recovery of the primal optimizer from the dual solution. Consider

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & \frac{1}{2}(x_1^2 + x_2^2 + x_3^2) \\ \text{s.t.} \quad & x_1 + x_2 + x_3 \leq -3. \end{aligned}$$

Lagrangian and dual function. Introducing a Lagrange multiplier $\lambda \geq 0$ for the inequality constraint, the Lagrangian is

$$L(x, \lambda) = \frac{1}{2}\|x\|^2 + \lambda(x_1 + x_2 + x_3 + 3).$$

The associated dual function is

$$d(\lambda) = \inf_{x \in \mathbb{R}^3} L(x, \lambda).$$

Minimization over the primal variables. Since the Lagrangian is strictly convex in x, the infimum is attained at the unique stationary point. Taking first-order conditions,

$$\frac{\partial L}{\partial x_i} = x_i + \lambda = 0, \qquad i = 1, 2, 3,$$

which yields

$$x_1 = x_2 = x_3 = -\lambda.$$

Dual function. Substituting this expression into the Lagrangian gives

$$\begin{aligned} d(\lambda) &= \frac{1}{2} \cdot 3\lambda^2 + \lambda(-3\lambda + 3) \\ &= -\frac{3}{2}\lambda^2 + 3\lambda. \end{aligned}$$

Dual maximization and primal recovery. The dual problem is

$$\max_{\lambda \geq 0} d(\lambda).$$

Since $d(\lambda)$ is a concave quadratic function, the maximizer satisfies

$$\frac{d}{d\lambda} d(\lambda) = -3\lambda + 3 = 0,$$

yielding $\lambda^\star = 1$. Recovering the primal solution from the minimizer of the Lagrangian, we obtain

$$x^\star = (-1, -1, -1),$$

which coincides with the solution obtained earlier and satisfies the constraint with equality.

5.3 Case Study: Duality in Linear Programming

We now specialize the general Lagrangian duality framework to linear programming and recover the classical LP dual problem as a special case.

Primal problem. Consider the linear program

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & c^\top x \\ \text{subject to} \quad & Ax \leq b, \\ & x \geq 0, \end{aligned} \tag{5.3}$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $c \in \mathbb{R}^n$ are given.

To place (5.7) into the standard convex minimization framework, we rewrite it equivalently as

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & -c^\top x \\ \text{subject to} \quad & Ax - b \leq 0, \\ & -x \leq 0. \end{aligned} \tag{5.4}$$

This is a convex optimization problem with linear objective and linear inequality constraints.

Lagrangian. Introduce Lagrange multipliers $\lambda_1 \in \mathbb{R}^m_+$ associated with the constraints $Ax \leq b$ and $\lambda_2 \in \mathbb{R}^n_+$ associated with the constraints $x \geq 0$. The Lagrangian is

$$\begin{aligned} \mathcal{L}(x, \lambda_1, \lambda_2) &= -c^\top x + \lambda_1^\top (Ax - b) - \lambda_2^\top x \\ &= \left(-c + A^\top \lambda_1 - \lambda_2\right)^\top x - \lambda_1^\top b. \end{aligned} \tag{5.5}$$

Dual function. The dual function is defined as

$$d(\lambda_1, \lambda_2) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda_1, \lambda_2).$$

From (5.5), the infimum over x is finite if and only if the linear coefficient of x vanishes, that is,

$$-c + A^\top \lambda_1 - \lambda_2 = 0.$$

If this condition is satisfied, then

$$d(\lambda_1, \lambda_2) = -\lambda_1^\top b;$$

otherwise, $d(\lambda_1, \lambda_2) = -\infty$.

Dual problem. The dual problem therefore becomes

$$\max_{\lambda_1 \geq 0,\, \lambda_2 \geq 0} d(\lambda_1, \lambda_2) \quad \text{subject to} \quad -c + A^\top \lambda_1 - \lambda_2 = 0.$$

Eliminating λ_2 yields the equivalent formulation

$$\begin{aligned} \min_{\lambda_1 \geq 0} \quad & \lambda_1^\top b \\ \text{subject to} \quad & A^\top \lambda_1 \geq c. \end{aligned} \tag{5.6}$$

This is precisely the classical dual linear program associated with (5.7).

Weak and strong duality for LP. For linear programs, weak duality holds automatically: for any primal feasible x and dual feasible λ_1,

$$c^\top x \leq \lambda_1^\top b.$$

Moreover, if either the primal or the dual problem is feasible and bounded, then *strong duality* holds:

$$\max_x c^\top x = \min_{\lambda_1 \geq 0} \lambda_1^\top b.$$

Complementary slackness. At optimality, primal–dual optimal solutions $(x^\star, \lambda_1^\star)$ satisfy the complementary slackness conditions

$$\lambda_{1,i}^\star (A_i x^\star - b_i) = 0, \quad i = 1, \ldots, m, \qquad x_j^\star (A_{\cdot j}^\top \lambda_1^\star - c_j) = 0, \quad j = 1, \ldots, n,$$

which fully characterize optimal solutions of linear programs.

Theorem 5.7 (Linear Programming Duality and Optimality)**.** *Consider the primal linear program*

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & c^\top x \\ \textit{subject to} \quad & Ax \leq b, \\ & x \geq 0, \end{aligned} \tag{5.7}$$

and its dual

$$\begin{aligned} \min_{\lambda \in \mathbb{R}^m} \quad & b^\top \lambda \\ \textit{subject to} \quad & A^\top \lambda \geq c, \\ & \lambda \geq 0. \end{aligned} \tag{5.8}$$

Then the following statements hold.

1. Weak duality. *For any primal feasible point x and any dual feasible multiplier λ, one has $c^\top x \leq b^\top \lambda$.*

2. Strong duality. *If either (5.7) or (5.8) is feasible and has a finite optimal value, then both problems admit optimal solutions and their optimal values coincide:*

$$\max\{c^\top x : Ax \leq b,\ x \geq 0\} = \min\{b^\top \lambda : A^\top \lambda \geq c,\ \lambda \geq 0\}.$$

3. Existence of optimal solutions. *If the primal problem (5.7) is feasible and bounded above, then (5.7) attains its maximum and (5.8) attains its minimum.*

4. Optimality characterization (complementary slackness). *A pair $(x^\star, \lambda^\star)$ is primal–dual optimal if and only if it is primal feasible and dual feasible, and satisfies the complementary slackness conditions*

$$\lambda_i^\star \, (b - Ax^\star)_i = 0, \quad i = 1, \ldots, m, \tag{5.9}$$

$$x_j^\star \, \big((A^\top \lambda^\star - c)_j\big) = 0, \quad j = 1, \ldots, n. \tag{5.10}$$

5. KKT specialization. *The conditions in item 4 are precisely the KKT conditions for linear programming; they are necessary and sufficient for optimality, without requiring differentiability or any constraint qualification beyond feasibility.*

Proof. Weak duality. Let x be primal feasible and λ be dual feasible. Since $A^\top \lambda \geq c$ and $x \geq 0$, we have

$$c^\top x \leq (A^\top \lambda)^\top x = \lambda^\top (Ax).$$

Since $Ax \leq b$ and $\lambda \geq 0$, it follows that

$$\lambda^\top (Ax) \leq \lambda^\top b = b^\top \lambda,$$

and hence $c^\top x \leq b^\top \lambda$.

Strong duality and existence. Linear programs are convex (indeed polyhedral) optimization problems. Assume that one of the two problems is feasible and has a finite optimal value. Then the feasible region of that problem is a nonempty polyhedron on which the objective is linear and bounded in the optimizing direction. It follows that the optimal value is attained at an extreme point of the feasible polyhedron. Moreover, by the linear programming strong duality theorem (equivalently, a separating hyperplane theorem/Farkas' lemma argument), the duality gap is zero: the primal and dual optimal values coincide, and an optimal solution exists for the other problem as well. This establishes items 2 and 3.

Optimality characterization. Assume first that $(x^\star, \lambda^\star)$ is primal–dual optimal. By weak duality,

$$c^\top x^\star \leq b^\top \lambda^\star,$$

and by strong duality the two sides are equal. Consider the nonnegative vector $b - Ax^\star \geq 0$ and the nonnegative vector $A^\top \lambda^\star - c \geq 0$. Using feasibility and the identity

$$b^\top \lambda^\star - c^\top x^\star = \lambda^{\star\top}(b - Ax^\star) + x^{\star\top}(A^\top \lambda^\star - c),$$

we obtain a sum of two nonnegative terms equal to zero, hence both terms must be zero:

$$\lambda^{\star\top}(b - Ax^\star) = 0, \qquad x^{\star\top}(A^\top \lambda^\star - c) = 0.$$

Since each term is a sum of nonnegative summands, each summand must be zero, which yields (5.9)–(5.10).

Conversely, suppose $x^\star$ and $\lambda^\star$ are feasible and satisfy (5.9)–(5.10). Then

$$b^\top \lambda^\star - c^\top x^\star = \lambda^{\star\top}(b - Ax^\star) + x^{\star\top}(A^\top \lambda^\star - c) = 0,$$

so $c^\top x^\star = b^\top \lambda^\star$. By weak duality, $c^\top x^\star$ is at most any dual feasible value and $b^\top \lambda^\star$ is at least any primal feasible value; hence both are optimal and $(x^\star, \lambda^\star)$ is primal–dual optimal.

KKT specialization. For linear programs, the Lagrangian stationarity condition coincides with $A^\top \lambda \geq c$ together with (5.10), and complementary slackness is exactly (5.9)–(5.10). Therefore, the conditions in item 4 are the KKT conditions, and by item 4 they are necessary and sufficient for optimality. □

5.4 Game-Theoretic Interpretation of Lagrangian Duality

A central object in constrained optimization is the *Lagrangian*, defined for the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \ldots, m, \end{aligned} \tag{5.11}$$

by

$$\mathcal{L}(x, \mu) \;:=\; f(x) + \sum_{i=1}^{m} \mu_i g_i(x), \qquad \mu \in \mathbb{R}^m_+.$$

A fundamental question concerns the relationship between the two optimization orders

$$\sup_{\mu \geq 0} \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \mu) \quad \text{and} \quad \inf_{x \in \mathbb{R}^n} \sup_{\mu \geq 0} \mathcal{L}(x, \mu).$$

Sup–inf formulation: the dual problem. The quantity

$$\sup_{\mu \geq 0} \inf_{x} \mathcal{L}(x, \mu)$$

defines the *dual problem*. For each fixed $\mu \geq 0$, the inner minimization over x yields the *dual function*

$$d(\mu) \;:=\; \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \mu). \tag{5.12}$$

The dual function is concave in μ, since it is the pointwise infimum of functions that are affine in μ. Maximizing $d(\mu)$ over $\mu \geq 0$ yields the tightest lower bound on the optimal value of the primal problem.

Inf–sup formulation: the primal problem. For any fixed $x \in \mathbb{R}^n$, we have

$$\sup_{\mu \geq 0} \mathcal{L}(x, \mu) = \begin{cases} f(x), & \text{if } g_i(x) \leq 0 \; \forall i, \\ +\infty, & \text{otherwise.} \end{cases}$$

Indeed, if any constraint $g_i(x) > 0$ is violated, the corresponding multiplier μ_i can be made arbitrarily large, driving the Lagrangian to $+\infty$.

Consequently,

$$\inf_{x} \sup_{\mu \geq 0} \mathcal{L}(x, \mu) = \inf\{f(x) \mid g_i(x) \leq 0, \; i = 1, \ldots, m\},$$

which coincides exactly with the original *primal problem* (5.11). Thus, the inf–sup formulation enforces feasibility by assigning infinite cost to constraint violations.

5.4.1 Saddle points and strong duality.

A pair $(x^\star, \mu^\star)$ is called a *saddle point* of the Lagrangian if

$$\mathcal{L}(x^\star, \mu) \;\le\; \mathcal{L}(x^\star, \mu^\star) \;\le\; \mathcal{L}(x, \mu^\star), \quad \forall x \in \mathbb{R}^n,\ \forall \mu \ge 0. \tag{5.13}$$

The existence of a saddle point implies equality of the two optimization orders:

$$\sup_{\mu \ge 0} \inf_x \mathcal{L}(x, \mu) = \inf_x \sup_{\mu \ge 0} \mathcal{L}(x, \mu),$$

and hence equality of the primal and dual optimal values.

Theorem 5.8 (Strong Duality and Saddle-Point Existence)**.** *Consider the convex optimization problem* (5.11)*, where f and g_i, $i = 1, \dots, m$, are convex functions. Suppose that* Slater's condition *holds, i.e., there exists a point $\bar{x} \in \mathbb{R}^n$ such that*

$$g_i(\bar{x}) < 0, \quad i = 1, \dots, m.$$

Then:

1. *strong duality holds:*
$$\inf_x \sup_{\mu \ge 0} \mathcal{L}(x, \mu) = \sup_{\mu \ge 0} \inf_x \mathcal{L}(x, \mu);$$
2. *there exists a saddle point $(x^\star, \mu^\star)$ of the Lagrangian satisfying* (5.13).

Proof. Let

$$p^\star := \inf\{f(x) \mid g_i(x) \le 0,\ i = 1, \dots, m\}$$

denote the primal optimal value, and define the dual function

$$d(\mu) := \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \mu), \qquad \mu \ge 0,$$

with dual optimal value

$$d^\star := \sup_{\mu \ge 0} d(\mu).$$

For any feasible x and any $\mu \geq 0$, we have

$$\mathcal{L}(x, \mu) = f(x) + \sum_{i=1}^{m} \mu_i g_i(x) \leq f(x),$$

since $g_i(x) \leq 0$ and $\mu_i \geq 0$. Taking the infimum over x and then the supremum over μ yields the weak duality bound

$$d^\star \leq p^\star.$$

To show the reverse inequality, define the set

$$\mathcal{C} := \left\{ \begin{bmatrix} f(x) - \alpha \\ g_1(x) \\ \vdots \\ g_m(x) \end{bmatrix} \middle| \; x \in \mathbb{R}^n, \; \alpha \in \mathbb{R} \right\} \subset \mathbb{R}^{m+1}.$$

Since f and g_i are convex, $\mathcal{C}$ is a convex set. By Slater's condition, there exists a strictly feasible point, which implies that $\mathcal{C}$ has nonempty interior.

For any $\alpha < p^\star$, the origin does not belong to $\mathcal{C}$, since otherwise there would exist x with $g_i(x) \leq 0$ and $f(x) < p^\star$, contradicting optimality of $p^\star$. Hence the closed convex set $\mathcal{C}$ and the point 0 can be strictly separated. By the separating hyperplane theorem, there exists a nonzero vector $(\lambda_0, \lambda) \in \mathbb{R}^{m+1}$ such that

$$\lambda_0(f(x) - \alpha) + \sum_{i=1}^{m} \lambda_i g_i(x) \geq 0 \quad \forall x, \; \forall \alpha < p^\star.$$

Necessarily $\lambda_0 > 0$, since otherwise the inequality would be violated by letting $\alpha \to -\infty$. Normalizing by λ_0 and defining $\mu_i := \lambda_i / \lambda_0 \geq 0$, we obtain

$$f(x) + \sum_{i=1}^{m} \mu_i g_i(x) \geq p^\star \quad \forall x.$$

Taking the infimum over x yields

$$d(\mu) \geq p^\star,$$

and therefore $d^\star \geq p^\star$. Combined with weak duality, this shows

$$d^\star = p^\star.$$

Since $d(\mu)$ is concave and upper semicontinuous and the feasible set $\{\mu \geq 0\}$ is closed and convex, the supremum defining $d^\star$ is attained at some $\mu^\star \geq 0$. Let $x^\star$ be a minimizer of $\mathcal{L}(\cdot, \mu^\star)$. Then

$$\mathcal{L}(x^\star, \mu^\star) = \inf_x \mathcal{L}(x, \mu^\star) = \sup_{\mu \geq 0} \inf_x \mathcal{L}(x, \mu).$$

For any $\mu \geq 0$, feasibility of $x^\star$ implies

$$\mathcal{L}(x^\star, \mu) \leq f(x^\star) = \mathcal{L}(x^\star, \mu^\star),$$

and by definition of $x^\star$,

$$\mathcal{L}(x^\star, \mu^\star) \leq \mathcal{L}(x, \mu^\star) \quad \forall x.$$

Hence $(x^\star, \mu^\star)$ is a saddle point of the Lagrangian. □

5.4.2 Game-Theoretic Interpretation and Minimax Theory

The Lagrangian formulation admits a natural interpretation as a two-player zero-sum game. The *primal player* selects a strategy $x \in \mathbb{R}^n$ with the objective of minimizing the payoff $\mathcal{L}(x, \mu)$, while the *dual player* selects a strategy $\mu \in \mathbb{R}^m_+$ with the objective of maximizing the same payoff. The value of the game therefore depends on the order in which the players act.

The quantity

$$\inf_{x \in \mathbb{R}^n} \sup_{\mu \geq 0} \mathcal{L}(x, \mu)$$

corresponds to the outcome when the primal player moves first and the dual player responds optimally. As shown earlier, this value coincides with the optimal value of the primal constrained optimization problem, since infeasible choices of x are penalized by arbitrarily large multipliers.

Conversely, the quantity

$$\sup_{\mu \geq 0} \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \mu)$$

corresponds to the outcome when the dual player moves first and the primal player responds optimally. This value defines the dual problem and provides the best lower bound on the primal optimal value via weak duality.

In general, for arbitrary payoff functions, these two values need not coincide. This asymmetry reflects the familiar phenomenon in game theory that the order of play may affect the

outcome. A fundamental question, therefore, is under what conditions the minimax equality

$$\inf_x \sup_{\mu \geq 0} \mathcal{L}(x, \mu) = \sup_{\mu \geq 0} \inf_x \mathcal{L}(x, \mu)$$

holds.

Classical minimax theorems in game theory, such as von Neumann's minimax theorem for finite games and Sion's minimax theorem for infinite-dimensional games, provide sufficient conditions for such an equality. These results typically require convexity (or linearity) in one player's strategy, concavity in the other's strategy, and appropriate compactness or interiority assumptions.

In the context of convex optimization, the Lagrangian $\mathcal{L}(x, \mu)$ is convex in x for fixed μ and affine (hence concave) in μ for fixed x. Slater's condition plays the role of a regularity assumption that substitutes for compactness and ensures that the feasible set has nonempty interior. Under these conditions, the hypotheses of a minimax theorem are effectively satisfied.

Strong duality can therefore be interpreted as a *minimax theorem* for the Lagrangian game: the order of play between the primal and dual players becomes immaterial, and the game admits a saddle-point equilibrium. That is, there exists a pair $(x^\star, \mu^\star)$ such that

$$\mathcal{L}(x^\star, \mu) \;\leq\; \mathcal{L}(x^\star, \mu^\star) \;\leq\; \mathcal{L}(x, \mu^\star), \quad \forall x,\ \forall \mu \geq 0.$$

From this perspective, the Karush–Kuhn–Tucker conditions characterize Nash equilibria of the associated zero-sum game. The primal variables represent the strategy of the minimizing player, the Lagrange multipliers represent the strategy of the maximizing player, and complementary slackness determines which constraints (or actions) are active at equilibrium.

5.5 Perturbations, Shadow Prices, and Sensitivity Analysis

We develop a unified view of *shadow prices* and *sensitivity analysis* by first establishing the perturbation theory for general convex optimization problems and then specializing the results to linear programs. This approach highlights linear-programming sensitivity formulas as direct consequences of convex duality.

5.5.1 Perturbations in Convex Optimization

Consider the convex optimization problem

$$\begin{aligned} \min_{x} \quad & f(x) \\ \text{s.t.} \quad & g_j(x) \leq 0, \quad j \in \mathcal{Q}, \end{aligned} \tag{5.14}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ and $g_j : \mathbb{R}^n \to \mathbb{R}$ are convex functions.

For $\theta \in \mathbb{R}^{|\mathcal{Q}|}$, define the perturbed optimal value function

$$\varphi(\theta) := \inf \left\{ f(x) \mid g_j(x) \leq \theta_j,\ j \in \mathcal{Q} \right\}, \tag{5.15}$$

so that the original problem corresponds to $\theta = 0$.

Assume that the unperturbed problem satisfies *Slater's condition*. Then strong duality holds, and there exists an optimal vector of KKT multipliers $\mu^\star = (\mu_j^\star)_{j \in \mathcal{Q}}$.

Fix an index $i \in \mathcal{Q}$ and consider perturbations of only the ith constraint,

$$\theta = \theta_i e^i, \qquad \theta_i \geq 0,$$

where e^i is the ith canonical basis vector.

Theorem 5.9 (Shadow-Price Interpretation of Multipliers)**.** *Under the above assumptions, the one-sided directional derivative of φ at $\theta = 0$ exists and satisfies*

$$\frac{d}{d\theta_i^+} \varphi(0) = \lim_{\theta_i \downarrow 0} \frac{\varphi(\theta_i e^i) - \varphi(0)}{\theta_i} = -\mu_i^\star. \tag{5.16}$$

Equivalently,

$$\mu_i^\star = -\frac{d}{d\theta_i^+} \varphi(0).$$

Proof. For $\theta = \theta_i e^i$, the perturbed problem takes the form

$$\varphi(\theta_i e^i) = \inf \left\{ f(x) \mid g_j(x) \leq 0,\ j \neq i,\ g_i(x) \leq \theta_i \right\}.$$

By Slater's condition for the unperturbed problem, strong duality holds in a neighborhood of $\theta = 0$. Hence, for all sufficiently small $\theta_i \geq 0$, the optimal value can be expressed in dual

form as

$$\varphi(\theta_i e^i) = \sup_{\mu \geq 0} \inf_x \Big\{ f(x) + \sum_{j \in \mathcal{Q}} \mu_j \big(g_j(x) - \theta_j \big) \Big\},$$

where $\theta_j = 0$ for $j \neq i$ and $\theta_i = \theta_i$.

Let $\mu^\star$ denote an optimal KKT multiplier vector for the unperturbed problem ($\theta = 0$). Evaluating the dual objective at $\mu^\star$ yields

$$\begin{aligned} \varphi(\theta_i e^i) &\geq \inf_x \Big\{ f(x) + \sum_{j \in \mathcal{Q}} \mu_j^\star g_j(x) - \mu_i^\star \theta_i \Big\} \\ &= \varphi(0) - \mu_i^\star \theta_i, \end{aligned}$$

where the equality follows from strong duality at $\theta = 0$.

Dividing both sides by $\theta_i > 0$ gives

$$\frac{\varphi(\theta_i e^i) - \varphi(0)}{\theta_i} \geq -\mu_i^\star. \tag{LB}$$

On the other hand, the function φ is convex in θ, since it is the pointwise infimum of affine functions of θ. Therefore, the one-sided directional derivative

$$\frac{d}{d\theta_i^+} \varphi(0) = \lim_{\theta_i \downarrow 0} \frac{\varphi(\theta_i e^i) - \varphi(0)}{\theta_i}$$

exists and equals the supremum of slopes of all affine functions supporting φ at $\theta = 0$.

Since the affine function

$$\theta_i \mapsto \varphi(0) - \mu_i^\star \theta_i$$

supports φ at $\theta = 0$, it follows that

$$\frac{\varphi(\theta_i e^i) - \varphi(0)}{\theta_i} \leq -\mu_i^\star \quad \text{for all sufficiently small } \theta_i > 0. \tag{UB}$$

Combining the lower bound (LB) and upper bound (UB), we conclude that

$$\lim_{\theta_i \downarrow 0} \frac{\varphi(\theta_i e^i) - \varphi(0)}{\theta_i} = -\mu_i^\star.$$

This establishes (5.16). The equivalent identity

$$\mu_i^\star = -\frac{d}{d\theta_i^+} \varphi(0)$$

follows immediately. □

The multiplier $\mu_i^\star$ therefore measures the *marginal decrease in optimal value* resulting from relaxing the ith constraint. For small perturbations,

$$\varphi(e^i) - \varphi(0) \approx -\mu_i^\star,$$

justifying the interpretation of $\mu_i^\star$ as a *shadow price*.

5.5.2 Linear Programming Case Study

We now recover classical linear-programming sensitivity results as direct corollaries of the convex perturbation framework.

Consider the linear program

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b, \\ & x \geq 0. \end{aligned} \tag{5.17}$$

Its dual is

$$\begin{aligned} \max_{u \in \mathbb{R}^m} \quad & b^\top u \\ \text{s.t.} \quad & A^\top u \leq c, \\ & u \geq 0. \end{aligned} \tag{5.18}$$

Assume feasibility and boundedness. Strong duality yields optimal solutions $(x^\star, u^\star)$ satisfying

$$c^\top x^\star = b^\top u^\star = f^\circ.$$

5.5.3 Sensitivity and Shadow-Price Formulas

Viewing the LP as a convex program with affine data, Theorem 5.9 immediately yields the classical sensitivity identities.

Theorem 5.10 (Shadow-Price Formulas for Linear Programs)**.** *Let $(x^\star, u^\star)$ be an optimal primal–*

dual pair. Then the optimal value f° satisfies

$$\frac{\partial f^\circ}{\partial c_i} = x_i^\star, \qquad i = 1, \ldots, n, \tag{5.19}$$

$$\frac{\partial f^\circ}{\partial b_j} = -u_j^\star, \qquad j = 1, \ldots, m, \tag{5.20}$$

$$\frac{\partial f^\circ}{\partial a_{ij}} = u_i^\star x_j^\star, \qquad i = 1, \ldots, m, \;\; j = 1, \ldots, n. \tag{5.21}$$

Proof. We derive each sensitivity formula by viewing the linear program as a special case of the convex perturbation framework established earlier.

Sensitivity with respect to b. Consider a perturbation of the right-hand side,

$$b \mapsto b + \theta e^j, \qquad \theta \geq 0,$$

where e^j is the jth canonical basis vector in $\mathbb{R}^m$. The perturbed primal problem becomes

$$\min_x \{c^\top x \mid Ax \leq b + \theta e^j, \; x \geq 0\}.$$

This is exactly of the form (5.15), with the constraint $A_j x \leq b_j$ relaxed by θ. By Theorem 5.9, the directional derivative of the optimal value satisfies

$$\frac{d}{d\theta^+} f^\circ = -u_j^\star,$$

where $u_j^\star$ is the optimal multiplier associated with the jth constraint. Hence,

$$\frac{\partial f^\circ}{\partial b_j} = -u_j^\star.$$

Sensitivity with respect to c. Now consider a perturbation of the objective coefficient c_i:

$$c \mapsto c + \theta e^i.$$

The perturbed objective is

$$(c + \theta e^i)^\top x = c^\top x + \theta x_i.$$

Let $x^\star$ be an optimal solution of the unperturbed problem. Since the feasible set is unchanged, we have

$$f^\circ(\theta) = \min_x \{c^\top x + \theta x_i\} \leq c^\top x^\star + \theta x_i^\star = f^\circ + \theta x_i^\star.$$

Conversely, optimality of $x^\star$ implies no feasible x can achieve a smaller first-order variation. Hence,

$$\lim_{\theta \downarrow 0} \frac{f^\circ(\theta) - f^\circ}{\theta} = x_i^\star,$$

which yields

$$\frac{\partial f^\circ}{\partial c_i} = x_i^\star.$$

Sensitivity with respect to A. Finally, perturb a single constraint coefficient a_{ij}:

$$a_{ij} \mapsto a_{ij} + \theta.$$

The ith constraint becomes

$$A_i x + \theta x_j \leq b_i.$$

Equivalently, this is a relaxation of the constraint by an amount θx_j. Applying the shadow-price result yields

$$\frac{d}{d\theta^+} f^\circ = -u_i^\star(-x_j^\star) = u_i^\star x_j^\star,$$

which gives

$$\frac{\partial f^\circ}{\partial a_{ij}} = u_i^\star x_j^\star.$$

Combining the three cases establishes (5.19)–(5.21). □

5.5.4 Economic Interpretation

- (5.19) shows that the optimal activity level $x_i^\star$ measures the sensitivity of the objective to changes in cost coefficients.
- (5.20) shows that the dual variable $u_j^\star$ equals the shadow price of resource j: relaxing the constraint reduces the objective at rate $u_j^\star$.
- (5.21) captures the joint effect of technology and resource valuation: changing how activity j consumes resource i affects the objective proportionally to both $u_i^\star$ and $x_j^\star$.

Thus, the familiar shadow prices of linear programming emerge as concrete instances of the general convex-analytic sensitivity theory.

6 Transformation of Nonconvex Problems into Convex Problems

Many optimization problems arising in control, learning, and engineering are *nonconvex* in their original formulation. Nevertheless, a broad and important class of such problems can be transformed into convex optimization problems through appropriate modeling techniques. This section presents several systematic approaches for transforming nonconvex problems into convex ones, either exactly (via equivalent reformulations) or approximately (via convex relaxations).

We consider optimization problems of the form

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad g_i(x) \leq 0, \quad i = 1, \ldots, m, \tag{6.1}$$

where the objective function $f : \mathbb{R}^n \to \mathbb{R}$ and the constraint functions $g_i : \mathbb{R}^n \to \mathbb{R}$ may be nonconvex.

Definition 6.1 (Equivalent reformulation)**.** A transformation of (6.1) is called an *equivalent reformulation* if it preserves the set of globally optimal solutions.

Definition 6.2 (Convex relaxation)**.** A transformation is called a *convex relaxation* if it produces a convex optimization problem whose feasible set contains that of the original problem. The optimal value of the relaxed problem provides a bound on the original one.

6.1 Epigraph Transformations

Definition 6.3 (Epigraph)**.** Given a function $f : \mathbb{R}^n \to \mathbb{R}$, its epigraph is defined as

$$\operatorname{epi}(f) := \{(x, t) \in \mathbb{R}^{n+1} \mid f(x) \leq t\}.$$

A fundamental result in convex analysis states that f is convex if and only if $\operatorname{epi}(f)$ is a convex set.

Basic epigraph reformulation. The unconstrained problem

$$\min_{x \in \mathbb{R}^n} f(x)$$

can be rewritten as

$$\min_{x,t} \; t \quad \text{subject to} \quad f(x) \leq t. \tag{6.2}$$

If f is convex, then (6.2) is a convex optimization problem equivalent to the original one.

Example 6.4 (Minimax objectives and epigraph reformulation)**.** Consider the optimization problem

$$\min_{x \in \mathbb{R}^n} \max_{j=1,\dots,p} g_j(x), \tag{6.3}$$

where each function $g_j : \mathbb{R}^n \to \mathbb{R}$ is convex.

Introducing an epigraph variable $t \in \mathbb{R}$, problem (6.3) can be equivalently rewritten as

$$\begin{aligned} \min_{x,t} \quad & t \\ \text{subject to} \quad & g_j(x) \le t, \quad j = 1, \dots, p. \end{aligned} \tag{6.4}$$

The reformulated problem is a convex optimization problem, since the objective is linear and each constraint defines a convex sublevel set. This epigraph representation is fundamental in robust optimization and worst-case control.

Example 6.5 (ℓ_∞-norm objectives and epigraph reformulation)**.** Consider the optimization problem

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|_\infty, \tag{6.5}$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

By definition of the infinity norm,

$$\|Ax - b\|_\infty = \max_{i=1,\dots,m} |(Ax - b)_i|.$$

Introducing an epigraph variable $t \in \mathbb{R}$, problem (6.5) is equivalent to

$$\begin{aligned} \min_{x,t} \quad & t \\ \text{subject to} \quad & -t \le (Ax - b)_i \le t, \quad i = 1, \dots, m. \end{aligned} \tag{6.6}$$

This formulation is a linear program, since both the objective and constraints are affine. The epigraph transformation converts the nonsmooth infinity-norm objective into a set of linear inequality constraints.

Example 6.6 (Maximum eigenvalue minimization)**.** Consider the optimization problem

$$\min_{x \in \mathbb{R}^n} \lambda_{\max}(A(x)), \tag{6.7}$$

where

$$A(x) := A_0 + \sum_{i=1}^{n} x_i A_i,$$

and $A_0, A_1, \ldots, A_n \in \mathbb{S}^m$ are given symmetric matrices. At first glance, problem (6.7) appears highly nonlinear due to the eigenvalue operator.

Introduce an epigraph variable $t \in \mathbb{R}$. Using the variational characterization of the maximum eigenvalue,

$$\lambda_{\max}(A) \leq t \quad \Longleftrightarrow \quad A \preceq tI,$$

problem (6.7) is equivalent to

$$\begin{aligned} \min_{x,t} \quad & t \\ \text{subject to} \quad & A_0 + \sum_{i=1}^{n} x_i A_i \preceq tI. \end{aligned} \tag{6.8}$$

The constraint

$$A_0 + \sum_{i=1}^{n} x_i A_i \preceq tI$$

is a linear matrix inequality (LMI) in the variables (x, t). The objective is linear. Therefore, problem (6.8) is a semidefinite program and hence a convex optimization problem.

Although the eigenvalue map $A \mapsto \lambda_{\max}(A)$ is nonlinear, it is a convex function on the space of symmetric matrices. The epigraph formulation exposes this convexity by replacing the eigenvalue inequality with an LMI. As a result, the original problem (6.7) admits an exact convex reformulation rather than a relaxation.

6.2 Change of Variables

A powerful method for transforming nonconvex optimization problems into convex ones is an appropriate *change of variables*. While the original formulation may obscure convexity, a reparameterization can expose an equivalent problem with a convex objective and convex constraints. This approach is particularly effective when the nonconvexity arises from multiplicative structure or ratios of functions.

Example 6.7 (Fractional objective and variable transformation)**.** Consider the optimization prob-

lem

$$\begin{aligned} \min_{x_1,x_2} \quad & \frac{2x_1+3x_2+2}{x_1+x_2+1} \\ \text{s.t.} \quad & A\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \le b, \end{aligned} \tag{6.9}$$

where $A \in \mathbb{R}^{m\times 2}$ and $b \in \mathbb{R}^m$ are given, and the denominator satisfies $x_1 + x_2 + 1 > 0$ on the feasible set.

Define the change of variables

$$y_1 := \frac{x_1}{x_1+x_2+1}, \qquad y_2 := \frac{x_2}{x_1+x_2+1}, \qquad z := \frac{1}{x_1+x_2+1}. \tag{6.10}$$

Under this transformation, we have the identity

$$x_i = \frac{y_i}{z}, \quad i = 1,2, \qquad x_1+x_2+1 = \frac{1}{z},$$

with $z > 0$.

Substituting into the objective function yields the linear expression

$$\frac{2x_1+3x_2+2}{x_1+x_2+1} = 2y_1 + 3y_2 + 2z.$$

The linear constraints transform as

$$A\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \le b \quad \Longleftrightarrow \quad A\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \le bz.$$

Therefore, problem (6.9) is equivalent to the linear optimization problem

$$\begin{aligned} \min_{(y_1,y_2,z)} \quad & 2y_1 + 3y_2 + 2z \\ \text{s.t.} \quad & A\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} - bz \le 0, \\ & z > 0. \end{aligned} \tag{6.11}$$

The feasible set of (6.11) is convex, and the objective function is linear.

The transformation in Example 6.7 converts a fractional program into an equivalent linear program by lifting the variables into a higher-dimensional space. This reformulation is exact:

every feasible solution of (6.9) corresponds uniquely to a feasible solution of (6.11), and vice versa. Such transformations form the basis of classical fractional programming methods and play an important role in convexification techniques.

Example 6.8 (Logarithmic change of variables and convexification)**.** Consider the optimization problem

$$\min_{d \in \mathbb{R}^n_{++}} \sum_{i,j} M_{ij} \frac{d_i^2}{d_j^2}, \tag{6.12}$$

where $M_{ij} \geq 0$ are given coefficients and $\mathbb{R}^n_{++} := \{d \in \mathbb{R}^n : d_i > 0 \; \forall i\}$.

Problem (6.12) is nonconvex in the variables d_i, due to the presence of multiplicative ratios. Introduce the logarithmic change of variables

$$x_i := \log d_i, \qquad i = 1, \ldots, n. \tag{6.13}$$

This transformation is bijective between $\mathbb{R}^n_{++}$ and $\mathbb{R}^n$.

Under (6.13), each ratio becomes $\frac{d_i^2}{d_j^2} = e^{2(x_i - x_j)}$. Hence, the objective function can be written as $\sum_{i,j} M_{ij} e^{2(x_i - x_j)}$. Taking the logarithm of the objective (which preserves minimizers since log is strictly increasing) yields the equivalent problem

$$\min_{x \in \mathbb{R}^n} \; f(x) := \log \left(\sum_{i,j} M_{ij} \, e^{2(x_i - x_j)} \right). \tag{6.14}$$

The function f in (6.14) is convex in x, since it is a log-sum-exp function of affine mappings of x.

6.2.1 Geometric Programming

Geometric programming (GP) is a class of optimization problems that admit an exact convex reformulation via a logarithmic change of variables. Although the original formulation is not convex, the transformed problem is convex and equivalent.

Monomials and posynomials. A *monomial* is a function $m : \mathbb{R}^n_{++} \to \mathbb{R}_{++}$ of the form

$$m(x) = c \prod_{i=1}^{n} x_i^{a_i},$$

where $c > 0$, $x_i > 0$, and $a_i \in \mathbb{R}$. A *posynomial* is a finite sum of monomials,

$$p(x) = \sum_{k=1}^{K} c_k \prod_{i=1}^{n} x_i^{a_{ik}}, \quad c_k > 0.$$

In general, posynomials are neither convex nor concave in the variable x.

Original geometric program. A geometric program in standard form is given by

$$\begin{aligned} \min_{x \in \mathbb{R}^n_{++}} \quad & p_0(x) \\ \text{subject to} \quad & p_i(x) \leq 1, \quad i = 1, \ldots, m, \\ & m_j(x) = 1, \quad j = 1, \ldots, r, \end{aligned} \tag{6.15}$$

where p_i are posynomials and m_j are monomials.

Logarithmic change of variables. Define the transformation

$$y := \log x, \quad \text{that is, } y_i = \log x_i, \ i = 1, \ldots, n.$$

This mapping is a bijection between $\mathbb{R}^n_{++}$ and $\mathbb{R}^n$, with inverse $x = \exp(y)$.

Transformed optimization problem. Applying the logarithmic transformation to the objective and constraints in (6.15), and taking logarithms of the objective and inequality constraints, yields the equivalent problem

$$\begin{aligned} \min_{y \in \mathbb{R}^n} \quad & \tilde{p}_0(y) := \log p_0(e^y) \\ \text{subject to} \quad & \tilde{p}_i(y) := \log p_i(e^y) \leq 0, \quad i = 1, \ldots, m, \\ & \tilde{m}_j(y) := \log m_j(e^y) = 0, \quad j = 1, \ldots, r. \end{aligned} \tag{6.16}$$

Convexity properties. For each monomial m_j,

$$\tilde{m}_j(y) = \log c_j + \sum_{i=1}^{n} a_{ij} y_i,$$

which is affine in y. For each posynomial p_i,

$$\tilde{p}_i(y) = \log \left(\sum_{k=1}^{K} \exp \left(\log c_{ik} + \sum_{l=1}^{n} a_{ilk} y_l \right) \right),$$

which is a log-sum-exp function and hence convex in y. Therefore, problem (6.16) is a convex optimization problem.

Because the logarithmic transformation is bijective, problems (6.15) and (6.16) are equivalent: an optimal solution $x^\star$ of the original problem corresponds to $y^\star = \log x^\star$ solving the transformed problem, and vice versa. Moreover, both problems have the same optimal value.

6.2.2 Fractional Programming

Fractional programming concerns optimization problems in which the objective function is expressed as a ratio of two functions. Such problems are generally nonconvex in their original formulation, even when the numerator is convex and the denominator is affine.

Original fractional optimization problem. Consider the optimization problem

$$\begin{aligned} \min_{x \in \mathcal{F}} \quad & \frac{f(x)}{g(x)} \\ \text{subject to} \quad & g(x) > 0, \end{aligned} \tag{6.17}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is a convex function, $g : \mathbb{R}^n \to \mathbb{R}_{++}$ is a positive affine function, and $\mathcal{F} \subseteq \mathbb{R}^n$ is a convex feasible set.

Affine denominator assumption. Assume that

$$g(x) = a^\top x + b, \quad a^\top x + b > 0 \ \ \forall x \in \mathcal{F}.$$

Charnes–Cooper transformation. Define the change of variables

$$y := \frac{x}{g(x)}, \qquad t := \frac{1}{g(x)}. \tag{6.18}$$

This transformation is bijective on the domain $g(x) > 0$, with inverse mapping

$$x = \frac{y}{t}, \quad t > 0.$$

Transformed optimization problem. Substituting $x = y/t$ into (6.17) yields the equivalent problem

$$\begin{aligned} \min_{y,t} \quad & t f\left(\frac{y}{t}\right) \\ \text{subject to} \quad & a^\top y + bt = 1, \\ & y \in t\mathcal{F}, \\ & t > 0, \end{aligned} \tag{6.19}$$

where $t\mathcal{F} := \{tx \mid x \in \mathcal{F}\}$.

Convexity properties. If f is convex, then the function

$$(y, t) \mapsto t f\left(\frac{y}{t}\right)$$

is the *perspective* of f and is jointly convex on the domain $t > 0$. The equality constraint $a^\top y + bt = 1$ is affine, and the set $t\mathcal{F}$ is convex whenever $\mathcal{F}$ is convex. Consequently, problem (6.19) is a convex optimization problem.

Equivalence. The Charnes–Cooper transformation establishes a one-to-one correspondence between feasible points of (6.17) and feasible points of (6.19). Moreover, the two problems have identical optimal values, and optimal solutions are mapped through $x^\star = y^\star / t^\star$.

6.2.3 Dinkelbach's Method for Fractional Programming

An alternative and widely used approach to fractional programming is *Dinkelbach's method*, which converts a fractional optimization problem into a sequence of parameterized convex optimization problems. In contrast to the Charnes–Cooper transformation, which produces a single equivalent convex reformulation, Dinkelbach's method yields an iterative procedure based on a fundamental equivalence between the original fractional problem and a family of difference-form problems.

Consider the fractional optimization problem

$$\min_{x \in \mathcal{F}} \quad \frac{f(x)}{g(x)}, \tag{6.20}$$

where $\mathcal{F} \subseteq \mathbb{R}^n$ is a nonempty compact convex set, $f : \mathbb{R}^n \to \mathbb{R}$ is a convex function, and $g : \mathbb{R}^n \to \mathbb{R}_{++}$ is a positive continuous function.

For each scalar parameter $\lambda \in \mathbb{R}$, define the auxiliary optimization problem

$$\phi(\lambda) \;:=\; \min_{x \in \mathcal{F}} \{f(x) - \lambda g(x)\}\,. \tag{6.21}$$

For fixed λ, problem (6.21) is a convex optimization problem whenever f is convex and g is affine or convex.

Equivalence and optimality characterization. Let

$$\lambda^\star := \min_{x \in \mathcal{F}} \frac{f(x)}{g(x)}.$$

Then the following statements hold:

- $\lambda^\star$ is the unique scalar satisfying

$$\phi(\lambda^\star) = 0. \tag{6.22}$$

- A point $x^\star \in \mathcal{F}$ is an optimal solution of (6.20) if and only if it minimizes (6.21) at $\lambda = \lambda^\star$.

Thus, solving the original fractional problem is equivalent to finding the unique root of the scalar function $\phi(\lambda)$.

Dinkelbach iteration. Motivated by the equivalence above, Dinkelbach's method generates a sequence $\{x^{(k)}, \lambda^{(k)}\}$ according to

$$x^{(k)} \in \arg\min_{x \in \mathcal{F}} \left\{f(x) - \lambda^{(k)} g(x)\right\}, \tag{6.23}$$

$$\lambda^{(k+1)} := \frac{f(x^{(k)})}{g(x^{(k)})}. \tag{6.24}$$

The iteration is terminated when $|f(x^{(k)}) - \lambda^{(k)} g(x^{(k)})|$ falls below a prescribed tolerance.

If f is convex, g is positive and affine, and $\mathcal{F}$ is compact, then the sequence $\{\lambda^{(k)}\}$ generated by Dinkelbach's method converges monotonically to the optimal value $\lambda^\star$. Under mild regularity conditions, the convergence is superlinear.

At each iteration, Dinkelbach's method requires the solution of a convex optimization problem of the form (6.21). As a result, the nonconvexity induced by the ratio structure is eliminated by replacing the fractional objective with a sequence of convex difference objectives.

Dinkelbach's method provides an exact solution approach for fractional programming without introducing auxiliary variables or lifted formulations. It is particularly advantageous when the feasible set $\mathcal{F}$ is complex or when variable transformations such as Charnes–Cooper are

undesirable.

6.3 Decomposition

Many large-scale optimization problems possess internal structure that can be exploited to simplify analysis and computation. A common theme is *decomposition*: rewriting a problem so that it can be solved by addressing smaller subproblems, either sequentially or in parallel. Decomposition plays a central role in distributed optimization, network resource allocation, and large-scale learning.

We distinguish between *primal (sequential) decomposition*, which eliminates variables directly, and *dual decomposition*, which relaxes coupling constraints via Lagrange multipliers.

6.3.1 Primal (Sequential) Decomposition

Example 6.9 (Sequential minimization)**.** Consider the optimization problem

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) \\ \text{s.t.} \quad & g(x_1) \leq 0, \\ & \tilde{g}(x_2) \leq 0. \end{aligned} \tag{6.25}$$

Define the *partial minimization* with respect to x_2:

$$\tilde{f}(x_1) := \min_{x_2} \; f(x_1, x_2) \quad \text{s.t.} \quad \tilde{g}(x_2) \leq 0. \tag{6.26}$$

Provided the minimum in (6.26) is attained for each x_1 of interest, the original problem (6.25) is equivalent to

$$\min_{x_1} \; \tilde{f}(x_1) \quad \text{s.t.} \quad g(x_1) \leq 0. \tag{6.27}$$

Thus, the original problem is reduced to a sequence of smaller problems: an inner optimization over x_2, followed by an outer optimization over x_1.

Remark 6.10. If f is jointly convex in (x_1, x_2) and $\tilde{g}$ defines a convex feasible set, then the partially minimized function $\tilde{f}$ is convex in x_1. Hence, primal decomposition preserves convexity.

6.3.2 Dual Decomposition

Primal decomposition is effective when variables can be separated directly. However, many problems are coupled through shared constraints. In such cases, *dual decomposition* provides a

powerful alternative by relaxing the coupling constraint using Lagrange multipliers.

Example 6.11 (Dual decomposition via a coupling constraint)**.** Consider the convex optimization problem

$$\begin{aligned} \min_{\{x_i\}_{i=1}^N} \quad & \sum_{i=1}^N f_i(x_i) \\ \text{s.t.} \quad & \sum_{i=1}^N x_i \leq C, \end{aligned} \tag{6.28}$$

where each $f_i : \mathbb{R} \to \mathbb{R}$ is convex and $C > 0$ is a shared resource constraint.

Introduce a Lagrange multiplier $\lambda \geq 0$ associated with the coupling constraint. The Lagrangian is

$$L(x_1, \ldots, x_N, \lambda) = \sum_{i=1}^N f_i(x_i) + \lambda \left(\sum_{i=1}^N x_i - C \right).$$

For a fixed λ, the Lagrangian separates as

$$L(x_1, \ldots, x_N, \lambda) = \sum_{i=1}^N \big(f_i(x_i) + \lambda x_i \big) - \lambda C.$$

The dual function is therefore

$$q(\lambda) = \sum_{i=1}^N \underbrace{\inf_{x_i} \big(f_i(x_i) + \lambda x_i \big)}_{\text{independent subproblem}} - \lambda C.$$

The dual problem is

$$\max_{\lambda \geq 0} \; q(\lambda). \tag{6.29}$$

Each subproblem involves only a single variable x_i and can be solved independently. Coordination between subsystems occurs solely through the shared dual variable λ.

Remark 6.12. Dual decomposition converts a globally coupled primal problem into a collection of decoupled subproblems linked by a low-dimensional dual variable. This structure underlies many distributed algorithms, including price-based resource allocation, congestion control, and network utility maximization.

Remark 6.13. When Slater's condition holds for (6.28), strong duality guarantees that solving the dual problem (6.29) recovers the primal optimum. In this case, primal and dual decomposition are exact, not relaxations.

6.4 Convex Relaxation

When an exact convex reformulation of a nonconvex optimization problem is not available, convexification may proceed through *relaxation*. In this approach, the original feasible set or objective is replaced by a convex superset or a convex underestimator. The resulting problem is convex but no longer equivalent to the original one; instead, it provides a bound or approximation to the original problem.

Convex relaxations play a central role in global optimization, approximation algorithms, and control, where exact solutions are often intractable.

6.4.1 Lifting and rank relaxation.

Quadratic and polynomial optimization problems can often be lifted to a higher-dimensional space in which objectives and constraints become linear. Nonconvexity is then isolated in rank constraints, which are subsequently dropped to obtain a convex relaxation.

Example 6.14 (Quadratically constrained quadratic program)**.** Consider the nonconvex problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & x^\top Q x \\ \text{subject to} \quad & x^\top A_i x \le b_i, \quad i = 1, \dots, m, \end{aligned} \tag{6.30}$$

where Q and A_i are symmetric but not necessarily positive semidefinite. It is a *quadratically constrained quadratic program* (QCQP). In general, such problems are nonconvex when either the objective matrix Q or one or more constraint matrices A_i fail to be positive semidefinite. As a consequence, the feasible set defined by the quadratic inequalities may be nonconvex, and the problem is, in general, NP-hard.

The lifting transformation $X = xx^\top$ maps the original vector variable $x \in \mathbb{R}^n$ into a matrix variable $X \in \mathbb{S}^n$, where $\mathbb{S}^n$ denotes the space of symmetric $n \times n$ matrices. Under this transformation, quadratic forms become linear:

$$x^\top Q x = \operatorname{tr}(Q x x^\top) = \operatorname{tr}(QX), \qquad x^\top A_i x = \operatorname{tr}(A_i X).$$

Thus, apart from the structural constraint $X = xx^\top$, the objective and constraints become affine functions of X.

The condition $X = xx^\top$ is equivalently characterized by the two constraints

$$X \succeq 0, \qquad \operatorname{rank}(X) = 1.$$

The positive semidefiniteness ensures that X can be factored as $X = YY^\top$ for some Y, while the rank-one constraint enforces that this factorization corresponds to a single vector x rather than a higher-dimensional embedding. It is precisely the rank constraint that renders the lifted formulation nonconvex.

The *Shor relaxation* is obtained by dropping the rank constraint and retaining only the convex condition $X \succeq 0$. The resulting problem is a semidefinite program (SDP):

$$\begin{aligned} \min_{X \succeq 0} \quad & \operatorname{tr}(QX) \\ \text{subject to} \quad & \operatorname{tr}(A_i X) \leq b_i, \quad i = 1, \ldots, m, \end{aligned}$$

which can be solved efficiently using interior-point or first-order methods.

Since the feasible set of the relaxed SDP strictly contains the set of rank-one matrices $xx^\top$, the optimal value of the relaxation provides a *lower bound* on the optimal value of the original QCQP. If the SDP solution happens to have rank one, then the relaxation is said to be *exact*, and the corresponding vector x can be recovered as a global optimizer of the original problem.

From a geometric perspective, the Shor relaxation replaces the nonconvex set

$$\{xx^\top : x \in \mathbb{R}^n\}$$

with its convex hull intersected with the positive semidefinite cone. This relaxation therefore plays a role analogous to convex envelopes in finite dimensions: it yields the tightest convex approximation obtainable through lifting and linearization of quadratic forms.

Shor relaxations are central to global optimization and approximation algorithms for nonconvex quadratic problems. They form the basis of polynomial-time approximation schemes in combinatorial optimization (e.g., Max-Cut), provide certificates of optimality gaps, and serve as bounding subproblems within branch-and-bound frameworks. In optimal control and signal processing, related relaxations arise in stability analysis, Lyapunov synthesis, and robust control via linear matrix inequalities.

Such rank relaxations form the basis of semidefinite approximation schemes for combinatorial optimization, polynomial optimization, and control synthesis.

6.4.2 Convex envelopes.

Given a nonconvex function defined on a compact domain, its *convex envelope* is the tightest convex function that underestimates it on that domain. Replacing the original function by its

convex envelope yields the strongest possible convex relaxation over the specified region.

Example 6.15 (Bilinear term relaxation)**.** The bilinear function $f(x, y) = xy$ is neither convex nor concave on a rectangular domain unless one of the variables is fixed. Geometrically, its graph over $[\underline{x}, \bar{x}] \times [\underline{y}, \bar{y}]$ is a saddle surface, and local optimality conditions alone are insufficient to characterize global behavior.

The *convex envelope* of f over a given domain is defined as the largest convex function that underestimates f everywhere on that domain. Equivalently, it is the pointwise supremum of all convex functions that lie below f. Replacing a nonconvex function by its convex envelope yields the tightest possible convex relaxation of the original problem.

For a bilinear term on a rectangular domain, the convex envelope admits a closed polyhedral representation. The key observation is that the graph of $f(x, y) = xy$ over the rectangle is completely characterized by its values at the four extreme points

$$(\underline{x}, \underline{y}),\ (\underline{x}, \bar{y}),\ (\bar{x}, \underline{y}),\ (\bar{x}, \bar{y}).$$

Each of the McCormick inequalities corresponds to an affine function whose graph is a supporting hyperplane to the bilinear surface at two of these corner points. Taken together, the four inequalities describe the convex hull of the set

$$\{(x, y, z) \in \mathbb{R}^3 : z = xy,\ x \in [\underline{x}, \bar{x}],\ y \in [\underline{y}, \bar{y}]\}.$$

Introducing an auxiliary variable $z = xy$, the nonlinear equality constraint $z = xy$ is replaced by the linear constraints

$$\begin{aligned} z &\geq \underline{x}y + x\underline{y} - \underline{x}\underline{y}, \\ z &\geq \bar{x}y + x\bar{y} - \bar{x}\bar{y}, \\ z &\leq \underline{x}y + x\bar{y} - \underline{x}\bar{y}, \\ z &\leq \bar{x}y + x\underline{y} - \bar{x}\underline{y}. \end{aligned}$$

The resulting feasible set in (x, y, z)-space is a polyhedron that coincides with the convex envelope of the bilinear graph over the rectangular domain.

This relaxation is exact at the boundary points of the rectangle and becomes progressively tighter as the variable bounds are tightened. In global optimization algorithms such as branch-and-bound and spatial branch-and-cut, the domain is recursively partitioned, and the corresponding McCormick relaxations are updated at each node. In the limit, as the domain shrinks

to a single point, the relaxation converges to the original bilinear constraint.

From a modeling perspective, McCormick relaxations provide a systematic mechanism for embedding nonconvex bilinear terms into mixed-integer linear or convex optimization frameworks, thereby enabling global solution methods while preserving tightness guarantees.

Convex envelopes are fundamental in branch-and-bound methods and mixed-integer convex programming.

6.4.3 Dual relaxations.

Lagrangian duality provides a systematic mechanism for convex relaxation by transferring nonconvexity from the primal variables to dual variables. Even when the primal problem is nonconvex, the dual problem is always convex and yields a lower bound on the primal optimal value.

Example 6.16 (Nonconvex quadratic minimization)**.** The optimization problem

$$\min_{x \in \mathbb{R}^n} \; x^\top Q x \quad \text{subject to} \quad \|x\|_2^2 \leq 1$$

is a special case of a *trust-region subproblem*. When the matrix Q is indefinite, the objective function is nonconvex and admits directions of negative curvature. As a result, classical convexity-based arguments do not apply, and local optimality conditions alone are insufficient to characterize the global solution.

Despite this nonconvexity, the problem possesses a remarkable hidden convex structure. To reveal it, consider the Lagrangian

$$\mathcal{L}(x, \lambda) = x^\top Q x + \lambda(x^\top x - 1), \qquad \lambda \geq 0.$$

For a fixed multiplier λ, the Lagrangian is a quadratic form in x:

$$\mathcal{L}(x, \lambda) = x^\top (Q + \lambda I) x - \lambda.$$

The dual function is obtained by minimizing $\mathcal{L}(x, \lambda)$ with respect to x. This minimum is finite if and only if

$$Q + \lambda I \succeq 0,$$

in which case the minimum is attained at $x = 0$ and equals $-\lambda$. If $Q + \lambda I$ is not positive

semidefinite, the quadratic form is unbounded below and the dual function equals $-\infty$.

The dual problem therefore takes the form

$$\begin{aligned} \max_{\lambda \geq 0} \quad & -\lambda \\ \text{subject to} \quad & Q + \lambda I \succeq 0. \end{aligned}$$

The constraint $Q + \lambda I \succeq 0$ is equivalent to

$$\lambda \geq -\lambda_{\min}(Q),$$

where $\lambda_{\min}(Q)$ denotes the smallest eigenvalue of Q. Hence the dual problem reduces to

$$\max_{\lambda \geq -\lambda_{\min}(Q)} (-\lambda),$$

whose optimal solution is

$$\lambda^\star = -\lambda_{\min}(Q), \qquad d^\star = \lambda_{\min}(Q).$$

On the primal side, observe that minimizing $x^\top Q x$ over the unit ball is equivalent to minimizing it over the unit sphere whenever Q has at least one negative eigenvalue. Indeed, along the eigenvector $v_{\min}$ associated with $\lambda_{\min}(Q)$,

$$x^\top Q x = \lambda_{\min}(Q)\|x\|^2,$$

and the minimum is achieved at $x = v_{\min}$ with $\|x\| = 1$. Therefore, the primal optimal value satisfies

$$p^\star = \lambda_{\min}(Q).$$

Since $p^\star = d^\star$, strong duality holds despite the nonconvexity of the primal problem. Moreover, the dual problem is convex: it consists of maximizing a linear function subject to a linear matrix inequality, or equivalently, minimizing the maximum eigenvalue of a shifted matrix. This example illustrates that certain nonconvex quadratic problems admit exact convex dual formulations, and that eigenvalue optimization naturally emerges as the dual counterpart of trust-region-type constraints.

From a systems and control perspective, this phenomenon underlies many classical results.

The appearance of eigenvalue constraints mirrors Lyapunov stability conditions, while the exactness of the relaxation explains why nonconvex energy minimization problems can often be solved reliably via convex optimization tools.

More generally, dual relaxations are exact for important problem classes such as trust-region subproblems and certain QCQPs, while providing tight bounds for broader classes of nonconvex problems.

Convex relaxations replace a difficult nonconvex feasible set with a convex set that contains it, or replace a nonconvex objective with a convex underestimator. The resulting solution provides a certificate, either as a bound or as an approximate solution, that can be refined or combined with other methods.

7 Hahn-Banach Theorems

The Hahn–Banach theorem provides the fundamental analytic mechanism underlying geometric separation results for convex sets. In this section, we make this connection precise by showing how convex sets give rise to sublinear functionals and how Hahn–Banach extensions yield separating hyperplanes.

Definition 7.1 (Sublinear functional)**.** Let X be a real vector space. A function $p : X \to \mathbb{R}$ is called *sublinear* if it satisfies:

1. *Positive homogeneity:*
$$p(\lambda x) = \lambda p(x), \quad \forall x \in X,\ \forall \lambda \geq 0,$$

2. *Subadditivity:*
$$p(x + y) \leq p(x) + p(y), \quad \forall x, y \in X.$$

Sublinear functionals generalize norms by allowing asymmetry and degeneracy. They measure size or cost in a way that is compatible with convex combinations and scaling, but do not necessarily arise from inner products.

Example 7.2 (Norms are sublinear)**.** Let $(X, \|\cdot\|)$ be a normed linear space. The norm function

$$p(x) := \|x\|$$

is sublinear. Positive homogeneity follows from the definition of a norm, and subadditivity is precisely the triangle inequality.

Example 7.3 (Maximum of linear functionals)**.** Let $X = \mathbb{R}^n$ and let $\{a_1, \ldots, a_m\} \subset \mathbb{R}^n$. Define

$$p(x) := \max_{1 \le i \le m} a_i^\top x.$$

Then p is sublinear. This follows from linearity of each map $x \mapsto a_i^\top x$ and the fact that the pointwise maximum of linear functionals is sublinear.

Example 7.4 (Support function of a convex set)**.** Let $K \subset \mathbb{R}^n$ be a nonempty compact convex set. Define

$$p(x) := \sup_{y \in K} x^\top y.$$

Then p is sublinear. This function is known as the *support function* of K and plays a central role in convex geometry and duality theory.

Why sublinear functionals encode convex sets. Let X be a real vector space and let $C \subset X$ be a nonempty convex set containing the origin. The geometry of C can be encoded by a function that measures how much C must be dilated in order to contain a given vector.

This leads to the *Minkowski functional* (or gauge) of C, defined by

$$p_C(x) := \inf\{\lambda > 0 : x \in \lambda C\}.$$

The function p_C assigns to each $x \in X$ the smallest scaling factor required to include x in the set C. Thus, $p_C(x) \le 1$ if and only if $x \in C$.

The structural properties of C are reflected directly in the algebraic properties of p_C. Convexity of C implies that p_C is subadditive, while containment of the origin implies positive homogeneity. Consequently, p_C is a sublinear functional.

Proof. $(1 \Rightarrow 2)$ Assume that p is sublinear. Let $x, y \in X$ and $\theta \in [0, 1]$. By subadditivity and positive homogeneity,

$$p(\theta x + (1-\theta)y) \le p(\theta x) + p((1-\theta)y) = \theta p(x) + (1-\theta)p(y),$$

which shows that p is convex. Positive homogeneity holds by assumption.

$(2 \Rightarrow 1)$ Assume that p is convex and positively homogeneous. For any $x, y \in X$, convexity with $\theta = \frac{1}{2}$ gives

$$p\left(\tfrac{x+y}{2}\right) \le \tfrac{1}{2}p(x) + \tfrac{1}{2}p(y).$$

Using positive homogeneity,

$$p(x+y) = 2p\left(\tfrac{x+y}{2}\right) \leq p(x) + p(y),$$

which establishes subadditivity. Positive homogeneity is given, hence p is sublinear. □

7.1 Equivalence Between Convex Sets and Sublinear Functionals

The relationship between convex geometry and sublinear analysis is made precise through the notion of the Minkowski functional. In this sense, convex sets containing the origin and sublinear functionals provide dual representations of the same structure.

Definition 7.5 (Minkowski functional)**.** Let X be a real vector space and let $C \subset X$ be a nonempty convex set containing the origin. The *Minkowski functional* (or *gauge*) of C is defined by

$$p_C(x) := \inf\{\lambda > 0 : x \in \lambda C\}, \qquad x \in X.$$

Proposition 7.6 (Gauge representation of convex sets)**.** *Let X be a real vector space and let $C \subset X$ be a convex set containing the origin. Then the Minkowski functional p_C is sublinear and satisfies*

$$C = \{x \in X : p_C(x) \leq 1\}.$$

Proof. Since C contains the origin and is convex, it is closed under convex combinations and positive scaling. Positive homogeneity of p_C follows directly from the definition. Subadditivity follows from convexity of C: if $x \in \lambda C$ and $y \in \mu C$, then

$$x + y \in (\lambda + \mu)C,$$

which implies $p_C(x+y) \leq \lambda + \mu$. Taking infima over all such λ, μ yields subadditivity.

Finally, by definition, $p_C(x) \leq 1$ if and only if $x \in C$, which establishes the set identity. □

Proposition 7.7 (Convex sets induced by sublinear functionals)**.** *Let $p : X \to \mathbb{R}$ be a sublinear functional. Define*

$$C_p := \{x \in X : p(x) \leq 1\}.$$

Then C_p is a convex set containing the origin.

Proof. Since $p(0) = 0$, the origin belongs to C_p. Convexity follows from subadditivity and positive homogeneity: for any $x, y \in C_p$ and $\theta \in [0,1]$,

$$p(\theta x + (1-\theta)y) \leq \theta p(x) + (1-\theta)p(y) \leq 1.$$

Thus $\theta x + (1-\theta)y \in C_p$. □

Propositions 7.6 and 7.7 establish a one-to-one correspondence between convex sets containing the origin and sublinear functionals, up to normalization. In this sense, convex geometry and sublinear analysis provide equivalent descriptions of the same underlying objects.

Theorem 7.8 (Equivalence between convex sets and sublinear functionals)**.** *Let X be a real vector space.*

1. *For every nonempty convex set $C \subset X$ containing the origin, the Minkowski functional p_C is sublinear and satisfies*

$$C = \{x \in X : p_C(x) \leq 1\}.$$

2. *For every sublinear functional $p : X \to \mathbb{R}$, the set*

$$C_p := \{x \in X : p(x) \leq 1\}$$

is a convex set containing the origin.

Moreover, the assignments

$$C \longmapsto p_C, \qquad p \longmapsto C_p$$

are mutually inverse up to normalization, in the sense that

$$p_{C_p} = p, \qquad C_{p_C} = C.$$

We now illustrate this correspondence with several canonical examples.

Example 7.9 (Euclidean unit ball and norm)**.** Let $X = \mathbb{R}^n$ and define

$$C := \{x \in \mathbb{R}^n : \|x\|_2 \leq 1\}.$$

Then C is a closed, convex, and centrally symmetric set containing the origin. Its Minkowski

functional is

$$p_C(x) = \inf\{\lambda > 0 : \|x\|_2 \leq \lambda\} = \|x\|_2.$$

Thus, the Euclidean norm is precisely the Minkowski functional of its unit ball. This example shows that norms arise as gauges of convex, balanced sets.

Example 7.10 (Asymmetric convex set and sublinear gauge)**.** Let $X = \mathbb{R}^2$ and define

$$C := \{(x, y) \in \mathbb{R}^2 : x \leq 1,\ |y| \leq 1\}.$$

The set C is convex and contains the origin, but is not symmetric about the origin. Its Minkowski functional is given by

$$p_C(x, y) = \inf\{\lambda > 0 : (x, y) \in \lambda C\} = \max\{x, |y|\}.$$

The function p_C is sublinear but not symmetric, and therefore does not define a norm. This example illustrates that sublinear functionals strictly generalize norms and faithfully encode asymmetries in the geometry of the underlying convex set.

Example 7.11 (Polyhedral gauge)**.** Let $X = \mathbb{R}^n$ and define

$$C := \{x \in \mathbb{R}^n : Ax \leq \mathbf{1}\},$$

where $A \in \mathbb{R}^{m \times n}$ and inequalities are interpreted componentwise. Then C is a convex polyhedron containing the origin. The associated Minkowski functional satisfies

$$p_C(x) = \inf\{\lambda > 0 : Ax \leq \lambda \mathbf{1}\} = \max_{1 \leq i \leq m} (A_i x),$$

where A_i denotes the ith row of A. Hence, p_C is the pointwise maximum of finitely many linear functionals, and is therefore sublinear.

7.2 Hahn–Banach Theorem (Extension Form)

Before stating the Hahn–Banach theorem, we present a concrete finite-dimensional example whose purpose is purely intuitive. In $\mathbb{R}^n$, every linear functional can be written explicitly as an inner product, which allows the extension guaranteed by Hahn–Banach to be visualized directly.

Example 7.12 (Explicit Hahn–Banach extension in $\mathbb{R}^2$)**.** Let $X = \mathbb{R}^2$ equipped with the Euclidean norm

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2}.$$

Define the sublinear functional

$$p(\mathbf{v}) := \|\mathbf{v}\|_2.$$

The sublevel set $\{\mathbf{v} \in \mathbb{R}^2 : p(\mathbf{v}) \leq 1\}$ is the closed unit disk.

Let

$$M := \{(\alpha, 0) : \alpha \in \mathbb{R}\}$$

be the x-axis, and define a linear functional

$$f : M \to \mathbb{R}, \qquad f(\alpha, 0) := \alpha.$$

For every $\mathbf{m} = (\alpha, 0) \in M$, we have

$$f(\mathbf{m}) = \alpha \leq |\alpha| = \|(\alpha, 0)\|_2 = p(\mathbf{m}),$$

so the domination condition $f \leq p$ holds on M.

Since X is finite-dimensional, every linear functional $F : \mathbb{R}^2 \to \mathbb{R}$ admits a representation as an inner product with a fixed vector. That is, there exists $\mathbf{y} \in \mathbb{R}^2$ such that

$$F(\mathbf{v}) = \mathbf{y}^\top \mathbf{v}, \qquad \forall \mathbf{v} \in \mathbb{R}^2.$$

The extension requirement $F|_M = f$ uniquely determines $\mathbf{y}$. Indeed, for all $\alpha \in \mathbb{R}$,

$$\mathbf{y}^\top (\alpha, 0) = \alpha,$$

which implies

$$\mathbf{y} = (1, 0).$$

Consequently, the extended functional is

$$F(x_1, x_2) = (1, 0)^\top (x_1, x_2) = x_1.$$

This functional satisfies:

1. $F|_M = f$, since $F(\alpha, 0) = \alpha$,

2. $F(\mathbf{v}) \leq \|\mathbf{v}\|_2$ for all $\mathbf{v} \in \mathbb{R}^2$, by the Cauchy–Schwarz inequality.

Geometric interpretation. The functional $F(\mathbf{v}) = x_1$ defines the supporting line

$$\{\mathbf{v} \in \mathbb{R}^2 : x_1 = 1\}$$

to the unit disk at the boundary point $(1, 0)$. This example illustrates explicitly how the Hahn–Banach theorem extends a linear functional defined on a subspace to a global supporting hyperplane in a finite-dimensional space.

We now state the general extension theorem.

Theorem 7.13 (Hahn–Banach Theorem)**.** *Let X be a real normed linear space and let $p : X \to \mathbb{R}$ be a continuous sublinear functional. Let $M \subset X$ be a linear subspace and $f : M \to \mathbb{R}$ a linear functional satisfying*

$$f(m) \leq p(m) \quad \forall m \in M.$$

Then there exists a linear functional $F : X \to \mathbb{R}$ extending f such that

$$F(x) \leq p(x) \quad \forall x \in X.$$

The Hahn–Banach theorem asserts that any linear inequality compatible with a convex (sublinear) constraint on a subspace can be extended globally. This extension principle is the analytic core from which supporting hyperplane and separation theorems for convex sets follow.

7.3 Separation of a Point and a Convex Set

We now show how Hahn–Banach yields a geometric separation theorem.

Theorem 7.14 (Separation by a hyperplane)**.** *Let X be a real normed linear space and let $C \subset X$ be a nonempty closed convex set. If $x_0 \notin C$, then there exists a continuous linear functional $F \in X^*$ and a real number α such that*

$$F(x_0) > \alpha \geq F(x) \quad \forall x \in C.$$

Proof. Define the translated set $C' := C - x_0$. Then C' is closed, convex, contains the origin, and $0 \notin C'$. Let $p := p_{C'}$ be the Minkowski functional of C'.

Define a linear functional f on the one-dimensional subspace $M := \operatorname{span}\{x_0\}$ by

$$f(\lambda x_0) := \lambda.$$

One verifies that $f(m) \leq p(m)$ for all $m \in M$.

By the Hahn–Banach theorem, f extends to a linear functional $F : X \to \mathbb{R}$ such that $F(x) \leq p(x)$ for all $x \in X$. Since $p(x) \leq 1$ for all $x \in C'$, it follows that

$$F(x) \leq 1 \quad \forall x \in C', \qquad F(x_0) = 1.$$

Translating back yields the desired separation. □

Consequently, in finite-dimensional convex optimization, every separating hyperplane takes the familiar form

$$\langle y, x \rangle = \alpha,$$

where $y \in \mathbb{R}^n$ is a normal vector to the hyperplane and $\alpha \in \mathbb{R}$ is a scalar offset.

Applying the separation theorem in $\mathbb{R}^n$, we obtain the following specialized result: if $C \subset \mathbb{R}^n$ is a nonempty closed convex set and $x_0 \notin C$, then there exist $y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$ such that

$$\langle y, x_0 \rangle > \alpha \geq \langle y, x \rangle \quad \forall x \in C.$$

The hyperplane $\{x \in \mathbb{R}^n : \langle y, x \rangle = \alpha\}$ separates the point x_0 from the convex set C, with y acting as the outward normal. Thus, in finite dimensions, the abstract Hahn–Banach separation theorem reduces to the existence of a supporting or separating hyperplane expressed by an inner product.

Strict versus non-strict separation. The strict inequality

$$F(x_0) > \alpha \geq F(x) \quad \forall x \in C$$

expresses a *strict separation* between the point x_0 and the convex set C. Whether strict separation is possible depends on the geometric and topological properties of C and the location of x_0.

If C is closed and convex and $x_0 \notin C$, then strict separation is always possible. Intuitively, closedness ensures that x_0 lies at a positive distance from C, allowing the separating hyperplane to be placed with a strict gap. Formally, the Hahn–Banach separation theorem guarantees the

existence of a functional F such that

$$\sup_{x \in C} F(x) < F(x_0).$$

If C is convex but not closed, or if x_0 lies in the closure $\overline{C}$, then strict separation may fail. In this case, one can generally only guarantee the existence of $F \in X^*$ and $\alpha \in \mathbb{R}$ such that

$$F(x_0) \geq \alpha \geq F(x) \quad \forall x \in C,$$

with equality possibly holding at x_0 or along a boundary sequence.

Geometrically, this corresponds to a supporting hyperplane that *touches* the closure of C at x_0 or asymptotically approaches it, leaving no room for a strict gap.

Boundary points. If $x_0 \in \partial C$, then separation is necessarily non-strict:

$$F(x_0) = \alpha \geq F(x) \quad \forall x \in C.$$

In this case, F defines a *supporting hyperplane* to C at the boundary point x_0.

7.4 Separation of Two Convex Sets

Once separation between a point and a convex set is established, the result can be extended to separation between two convex sets. This generalization captures the fundamental geometric principle underlying duality and feasibility certification in convex analysis.

Theorem 7.15 asserts that two disjoint convex sets can be separated by a hyperplane. The linear functional F defines a hyperplane

$$\{x \in X : F(x) = \alpha\}$$

such that C_1 lies entirely on one side of the hyperplane and C_2 lies on the other. This is referred to as *weak (or non-strict) separation*, since equality may occur for boundary points of the sets.

Theorem 7.15 (Separation of two convex sets)**.** *Let X be a real normed linear space and let $C_1, C_2 \subset X$ be nonempty convex sets with $C_1 \cap C_2 = \varnothing$. Then there exists a nonzero continuous linear functional*

$F \in X^*$ and a scalar $\alpha \in \mathbb{R}$ such that

$$F(x) \leq \alpha \leq F(y) \quad \forall x \in C_1,\ \forall y \in C_2.$$

The separation of a point from a convex set is a special case of Theorem 7.15. Indeed, by taking $C_2 = \{x_0\}$ and $C_1 = C$, the condition $C_1 \cap C_2 = \varnothing$ reduces to $x_0 \notin C$, and the conclusion recovers the point–set separation theorem.

A standard proof reduces the problem to separating the origin from a convex set. Define the difference set

$$C := C_1 - C_2 = \{x - y : x \in C_1,\ y \in C_2\}.$$

The set C is convex and does not contain the origin. Applying the point–set separation theorem to 0 and C yields a linear functional whose inequality rearranges to the stated separation between C_1 and C_2.

Strict separation. In general, the inequalities in Theorem 7.15 cannot be made strict. However, if additional regularity assumptions hold, such as one set being compact and the other closed, then there exist $\alpha_1 < \alpha_2$ such that

$$F(x) \leq \alpha_1 < \alpha_2 \leq F(y) \quad \forall x \in C_1,\ \forall y \in C_2.$$

This corresponds to a hyperplane separating the two sets with a positive gap.

Finite-dimensional specialization. If $X = \mathbb{R}^n$, then every continuous linear functional $F \in X^*$ can be written as

$$F(x) = \langle y, x \rangle$$

for some vector $y \in \mathbb{R}^n$. In this case, separation is expressed by a hyperplane of the form

$$\langle y, x \rangle = \alpha,$$

with y acting as the normal direction.

8 Further Reading

The material in this chapter draws on classical results in convex analysis, duality theory, and optimization, as well as important application-driven developments in communications and control. The following references provide authoritative treatments and complementary perspectives.

Saddle-point theory provides the unifying framework behind Lagrangian duality, KKT conditions, and constrained optimization. A classical and rigorous treatment can be found in [3], which develops saddle-point optimality conditions, constraint qualifications, and duality in both convex and nonconvex settings. Closely related geometric and variational perspectives are presented in [5, 7]. For a modern, optimization-focused exposition emphasizing duality and KKT conditions, see [1].

The water-filling solution is a canonical example illustrating dual variables, complementary slackness, and optimal resource allocation. It appears prominently in information theory and communications. Classical derivations are given in [2]. Convex-optimization interpretations of water-filling and its generalizations to network utility maximization are discussed in [1] and in the broader literature on network optimization.

Fractional programming problems arise naturally in energy efficiency, throughput optimization, and economic modeling. The foundational theory, including the Charnes–Cooper transformation and Dinkelbach's method, is developed in the monograph [6, 9]. Algorithmic and parametric perspectives on fractional programming are also discussed in the literature on quasiconvex optimization and parametric programming.

Duality provides a systematic mechanism for convex relaxation of nonconvex problems, including quadratic and polynomial optimization. The role of duality in trust-region methods and quadratic programming is discussed in [4]. Semidefinite relaxation and duality-based approximation methods are treated in depth in [8]. These ideas also form the foundation of sum-of-squares and moment-based relaxations in polynomial optimization.

9 Exercises

Exercise 1 (Convex Sets)**.**

(i) Let $C \subset \mathbb{R}^n$ be a convex set, and let $x_1, \ldots, x_k \in C$. Suppose $\theta_1, \ldots, \theta_k \in \mathbb{R}$ satisfy

$$\theta_i > 0, \qquad \theta_1 + \cdots + \theta_k = 1.$$

Show that

$$\theta_1 x_1 + \cdots + \theta_k x_k \in C.$$

(ii) Let $C \subset \mathbb{R}^n$ be the solution set of the quadratic inequality

$$C = \{x \in \mathbb{R}^n \mid x^\top A x + b^\top x + c \leq 0\},$$

where $A \in \mathbb{R}^{n \times n}$, $b \in \mathbb{R}^n$, and $c \in \mathbb{R}$. Show that C is convex if A is positive semidefinite.

(iii) Show that the hyperbolic set

$$\{x \in \mathbb{R}^2_+ \mid x_1 x_2 \geq 1\}$$

is convex. As a generalization, show that the set

$$\left\{x \in \mathbb{R}^n_+ \;\middle|\; \prod_{i=1}^n x_i \geq 1\right\}$$

is convex.

Exercise 2 (Convex Functions)**.**

(i) Let $g : \mathbb{R}^n \to \mathbb{R}$ be a convex function and let $f : \mathbb{R} \to \mathbb{R}$ be an increasing convex function. Show that the composition $f(g(x))$ is a convex function.

(ii) Show that the log-sum-exp function $f : \mathbb{R}^n \to \mathbb{R}$ defined by

$$f(x) = \log\left(\sum_{i=1}^n e^{x_i}\right)$$

is convex.

(iii) Let $p < 1$, $p \neq 0$. Show that the function $f : \mathbb{R}^n_{++} \to \mathbb{R}$ defined by

$$f(x) = \left(\sum_{i=1}^n x_i^p\right)^{1/p}$$

is concave.

(iv) Show that the function $f : \mathbb{R}^n \to \mathbb{R}$ defined by

$$f(x) = \|Ax - b\|,$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $\|\cdot\|$ is a norm on $\mathbb{R}^m$, is convex.

Exercise 3 (Integral Averaging of a Convex Function)**.** Let $\Phi : [0, \infty) \to \mathbb{R}$ be a convex function, and define

$$f(t) := \frac{1}{t} \int_0^t \Phi(\alpha)\, d\alpha, \qquad t > 0.$$

1. Show that f is well defined for all $t > 0$.
2. Prove that f is convex on the interval $(0, \infty)$.

Exercise 4 (Failure of Convexity via the Hessian)**.** Consider the function

$$f(\alpha, \beta) = \alpha^\alpha + \beta^\beta - \alpha^\beta - \beta^\alpha, \qquad \alpha > 0,\ \beta > 0.$$

1. Compute the second-order partial derivative $\frac{\partial^2 f}{\partial \alpha^2}$.
2. Show that
$$\frac{\partial^2 f}{\partial \alpha^2}(1, e) = 2 - e^2 < 0.$$
3. Conclude that the Hessian matrix of f is not positive semidefinite at $(\alpha, \beta) = (1, e)$.
4. Deduce that f is not convex in any neighborhood of $(1, e)$, and hence is not convex in the interior of the first quadrant.

Exercise 5 (Infimal Convolution and Convexity)**.** Consider the function $g : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ defined by

$$g(x, y) = \|y\|_2 + \frac{1}{2}\|x - y\|_2^2.$$

(i) Find

$$f(x) := \inf_{y \in \mathbb{R}^n} g(x, y).$$

(ii) Show that the function f is convex on $\mathbb{R}^n$.

Exercise 6 (Log-Sum-Exp Optimization and Duality)**.** Let $x, a_i \in \mathbb{R}^n$ for $i = 1, \ldots, m$, and let $b_i, y_i \in \mathbb{R}$. Let $p_i \in (0, 1]$, $i = 1, \ldots, m$, be weighting factors. Consider the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \log\left(\sum_{i=1}^m p_i e^{y_i}\right) \\ \text{s.t.} \quad & a_i^\top x + b_i = y_i, \quad i = 1, \ldots, m. \end{aligned}$$

(i) Show that the above problem is a convex optimization problem.

(ii) Derive the dual problem.

(iii) Determine whether the dual problem is concave, and justify your answer.

Exercise 7 (KKT Conditions)**.** Use the Karush–Kuhn–Tucker (KKT) conditions to solve the following optimization problem and determine the optimal solution $(x_0, x_1, \ldots, x_n)$:

$$\begin{aligned} \max_{x_0, x_1, \ldots, x_n} \quad & \log(1 + x_0) + \sum_{i=1}^{n} \log x_i \\ \text{s.t.} \quad & x_0 + x_i \leq 1, \quad i = 1, \ldots, n, \\ & x_0, x_1, \ldots, x_n \geq 0. \end{aligned}$$

Exercise 8 (Entropy-Regularized Transport Duality)**.** Let $\Sigma = \{\Sigma_{ij} \mid i = 1, \ldots, n,\ j = 1, \ldots, m\}$ and $C = \{C_{ij} \mid i = 1, \ldots, n,\ j = 1, \ldots, m\}$ be real $n \times m$ matrices. Define the inner product

$$\langle \Sigma, C \rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} C_{ij} \Sigma_{ij}.$$

Let $a \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, and denote by $\mathbf{1}_n$ and $\mathbf{1}_m$ the vectors of all ones in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively.

Consider the optimization problem

$$\begin{aligned} \text{(P1)} \qquad \min_{\Sigma} \quad & \langle \Sigma, C \rangle + \lambda \sum_{i=1}^{n} \sum_{j=1}^{m} \Sigma_{ij} \log \Sigma_{ij} \\ \text{s.t.} \quad & \Sigma \mathbf{1}_m = a, \\ & \Sigma^\top \mathbf{1}_n = b, \\ & \Sigma_{ij} \geq 0, \quad i = 1, \ldots, n,\ j = 1, \ldots, m, \end{aligned}$$

where $\lambda > 0$. Derive the dual problem associated with (P1).

Exercise 9 (Quasi-Concavity)**.** Let $f : C \subset \mathbb{R}^n \to \mathbb{R}$ be a function defined on a convex set C. The function f is said to be *quasi-concave* if

$$f(\lambda x_1 + (1 - \lambda) x_2) \geq \min\{f(x_1), f(x_2)\}, \quad \forall x_1, x_2 \in C,\ \forall \lambda \in [0, 1].$$

(i) Show that every concave function is quasi-concave.

(ii) Let $f : C \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable. Show that f is quasi-concave if and only if

$$f(x_2) \leq f(x_1) \quad \Longrightarrow \quad \nabla f(x_2)^\top (x_1 - x_2) \geq 0, \quad \forall x_1, x_2 \in C.$$

Exercise 10 (Bregman Divergence and Mirror Projections)**.** Let $\mathcal{X}$ be a closed convex set and

let $f : \mathcal{X} \to \mathbb{R}$ be a continuously differentiable convex function. The Bregman divergence associated with f is defined by

$$D_f(x, y) = f(x) - f(y) - \langle \nabla f(y), x - y \rangle.$$

An example is the negative entropy

$$f(x) = \sum_{i=1}^{n} x_i \log x_i,$$

where $x = (x_1, \ldots, x_n)$ belongs to the n-dimensional probability simplex.

The Bregman projection of y onto $\mathcal{X}$ is defined as

$$\Pi_{\mathcal{X}}^{f}(y) = \arg\min_{x \in \mathcal{X}} D_f(x, y).$$

(i) Let $\hat{x} = \nabla f(x)$ and $\hat{y} = \nabla f(y)$. Let f^* denote the Fenchel conjugate of f. Show that

$$\nabla f^*(\hat{x}) = x, \qquad D_f(x, y) = D_{f^*}(\hat{y}, \hat{x}).$$

(ii) Show that for all $x, y, z \in \mathcal{X}$,

$$D_f(x, y) + D_f(z, x) - D_f(z, y) = (\nabla f(x) - \nabla f(y))^\top (x - z).$$

(iii) Assume that f is strictly convex. Let $\pi = \Pi_{\mathcal{X}}^{f}(y)$. Show that

$$(\nabla f(y) - \nabla f(\pi))^\top (w - \pi) \leq 0, \quad \forall w \in \mathcal{X}.$$

(iv) Let $\mathcal{X}$ be the probability simplex and let D be a convex set such that $\mathcal{X} \cap D \neq \emptyset$. For $y \in D \setminus \mathcal{X}$, show that the Bregman projection under negative entropy satisfies

$$\Pi_{\mathcal{X} \cap D}^{f}(y) = \frac{y}{\|y\|_1}.$$

Exercise 11 (Lasso Problem and Its Dual (10 pts))**.** Consider the standard Lasso problem

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|y - Ax\|_2^2 + \lambda \|x\|_1,$$

where $x, y \in \mathbb{R}^n$ and $\lambda \geq 0$. This unconstrained problem can be equivalently rewritten as the constrained optimization problem

$$\begin{aligned} \min_{u,x} \quad & \frac{1}{2}\|y-u\|_2^2 + \lambda\|x\|_1 \\ \text{s.t.} \quad & u = Ax. \end{aligned}$$

(i) Show that the Lagrangian dual of the reformulated problem is given by

$$\begin{aligned} \min_{r \in \mathbb{R}^n} \quad & \frac{1}{2}\|y-r\|_2^2 \\ \text{s.t.} \quad & \|A^\top r\|_\infty \leq \lambda. \end{aligned}$$

(ii) Define the set

$$C := \{z \in \mathbb{R}^n \mid \|A^\top z\|_\infty \leq \lambda\},$$

and let $\Pi_C : \mathbb{R}^n \to C$ denote the Euclidean projection onto C. Show that the optimal primal and dual solutions satisfy

$$u^\star = y - \Pi_C(y), \qquad x^\star = A^\dagger u^\star,$$

where $A^\dagger$ denotes the Moore–Penrose pseudoinverse of A.

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